



6G AI-Native Integrated RAN-Core Networks

Deliverable D4.1

State of the art on AI/ML Solutions for RAN and Core Optimization



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Abstract

This deliverable presents the state of the art on AI/ML solutions for RAN and Core optimization in the context of the 6GARROW project. It supports Objective 3: Optimize RAN and Core based on AI/ML solutions, and constitutes the first measurable result toward this objective. The document reviews existing approaches and outlines planned studies on three main themes: (i) AI-native and semantic 6G architectures, (ii) AI-driven resource usage and energy optimization, and (iii) automation and failure recovery across RAN and Core domains. The analysis establishes a baseline for future 6GARROW work, guiding subsequent research, integration, and validation activities toward trustworthy, energy-efficient, and resilient AI-native 6G networks.

Keywords

6G, AI-native networks, Machine Learning, RAN optimization, Core network, Automation, Resilience, Semantic communication, Energy efficiency, Network architecture, EU AI Act, Digital twin, Resource management, Cross-domain coordination, State of the art

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Executive Summary

Deliverable D4.1 establishes the baseline for the research and innovation activities of the 6GARROW project toward AI-native, energy-aware and resilient 6G networks. It directly supports Objective 3: Optimize RAN and Core based on AI/ML solutions, by reviewing the current state of the art and defining the initial research directions for integrating AI across the radio access and core network domains.

This deliverable is structured around three thematic axes that reflect the scientific ambitions and measurable results of the project, namely:

1. Architecture & Enablers, which investigates the evolution of 6G architectures toward semantic, AI-native, and regulation-compliant designs, including compliance with the forthcoming EU AI Act;
2. AI-Native 6G Resource Usage, focusing on AI/ML-driven algorithms for the optimization of energy efficiency and spectrum utilization at both network and physical layers; and
3. Automation and Failure Recovery, addressing proactive resilience, and cross-domain AI coordination mechanisms for distributed 6G systems.

For each topic, the deliverable outlines the planned research studies of the consortium partners and provides a comprehensive state-of-the-art survey positioning these studies within ongoing 5G-Advanced and forthcoming 6G research initiatives. The outcome is a consolidated view of the current technological landscape and the research gaps that 6GARROW will address in subsequent activities (D4.2 and D4.3).

Abbreviations

3GPP	third generation partnership project
5G	5 th generation
5GC	5G core
5QI	5G quality indicator
6G	6 th generation
ADRF	analytics data repository function
AF	application function
AI	artificial intelligence
AODT	aerial omniverse digital twin
API	application programming interface
ASM	advanced sleep mode
CAPEX	capital expenditure
CAPIF	common API framework
CN	core network
CSI	channel state information
DCCF	data collection coordination function
DDQN	double deep Q-network
DQN	deep Q-network
DRL	deep reinforcement learning
DT	digital twin
E2E	end to end
eMBB	enhanced mobile broadband
ER	expected result
EU	European union
FL	federated learning
GA	genetic algorithms
GAN	generative adversarial network
GPS	Global positioning system
ISAC	integrated sensing and communication
JCAS	joint communications and sensing
KPI	key performance indicator
LiDAR	light detection and ranging
LLM	large language model
LLM4CP	LLM-empowered channel prediction method
LSTM	long short-term memory
MDA	management data analytics
MDAF	management analytics function
MDAS	management data analytics service
MIMO	multiple input multiple output
ML	machine learning
MME	mobility management entity
MNO	mobile network operator
NEF	network exposure function
NETCONF	network configuration protocol
NextG	next generation
NF	network function
NOMA	non-orthogonal multiple access
Non-RT	non-real-time
NRF	network repository function
NSM	network slice management

NWDAF	network data analytics function
O&M	Operation and maintenance
OAM	operations, administration and management
OFDMA	orthogonal frequency division multiple access
OPEX	operations expenditure
O-RAN	open radio access network
PCF	policy control function
PHY	physical
PLMN	public land mobile network
PPO	proximal policy optimization
PRB	physical layer block
PUE	power usage effectiveness
QoE	quality of experience
QoS	quality of service
RAN	radio access network
RIC	RAN intelligent controller
RIS	reconfigurable intelligent surface
RL	Reinforcement learning
RRM	radio resource management
RSMA	rate-splitting multiple access
RTT	round trip time
RU	radio unit
SBA	service based architecture
SBA	service-based architecture
SBI	service-based interface
SBS	small base stations
SDN	software define networks
SEPP	security edge protection proxy
SFL	split federated learning
SINR	signal-to-Interference-plus-noise ratio
SL	split learning
SMO	service management and orchestration
SNPN	standalone non-public networks
SRS	sounding reference signal
SSRF	semantic state representation function
UE	User equipment
UPF	user plane function
URLLC	ultra-reliable low-latency communications
VM	Virtual machine
XR	extended reality
ZSM	zero-touch network and service management

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1. Introduction

The 6GARROW project sets out a bold vision for the future of communication networks, which is related to the design a truly artificial intelligence (AI)-native and energy-aware resilient 6G architecture with the ability of adapting dynamically, operating sustainably, and providing trustworthy automation. As stated in the project objectives, the ambition is to embed AI deeply into the network fabric, enabling self-optimization, proactive failure recovery, and semantic service provisioning across radio access network (RAN) and core domains.

Deliverable D4.1 directly contributes to this vision by establishing a foundation of knowledge on the state of the art in AI-driven architectures, resource usage, and automation mechanisms for 6G. In addition, it outlines the avenues of research that the consortium partners are pursuing to address the gaps identified. This dual perspective, which includes both planned work and existing knowledge, ensures that the activities of the project are both innovative and grounded in the current technological landscape.

This deliverable addresses 6GARROW's Objective 3 as shown in the table below. It serves as a baseline for subsequent experimental, integration, and validation activities.

Objective 3 Optimize RAN and Core based on AI/ML Solutions	
Expected Results (ER)	ER3.1: Energy efficient AI/ML algorithms for overall RAN and core optimization and automation that can exploit the network's AI nativeness to efficiently orchestrate and scale network functions and resources in distributed architectures under energy consumption constraints. ER3.2: Semantic and goal-oriented radio resource management techniques, including coexistence between empowered and non-empowered users. ER3.3: AI-driven platform for automated base station management, aiming to optimize network performance and efficiency in real-time. ER3.4: AI/ML service provision intra-network and externally, enabling consumers to access intra-network data and AI training capabilities for tailored AI models. ER3.5: Intelligent robustness and recovery mechanisms for CN functions in distributed architectures under QoS requirements. ER3.6: RAN-Core integrated analytics and learning model management technology for cross-domain inference coordination which are geared to be exposed to network automation entities of each domain.
Pertinence to the HORIZON-JU-SNS-2024-STREAM-B-01-06 work programme	Relates to Expected Outcome “EU-ROK research collaboration in targeting RAN and integrated device-network approaches, evolving through 6G standardisation discussions and paving the way for future advancements towards AI-native RANs.” and Scope “Procedures and protocols empowered by AI that improve efficiencies of the wireless communications through mobility management, wireless resource management, automated maintenance, and self-optimization of network parameters.”
Measurable results	Comprehensive analytical proofs or simulation-based demonstrations that conclusively establish the effectiveness and efficiency of the proposed innovative algorithms and methods in achieving the ERs.
Means of verification	D4.1, D4.2, D4.3. Publications

The document is organized into three main parts that directly map onto the ambition of the project:

- Architecture & Enablers (Section 2): addressing the integration of semantic and AI-driven intelligence into network design, in line with the ambition to create AI-native architectures that comply with trust and transparency requirements such as the EU AI Act.
- AI Native 6G Resource Usage (Section 3): exploring AI techniques at the physical and network layers to improve energy efficiency and spectrum usage, aligning with the project's measurable key performance indicators (KPIs) on energy reduction.
- Automation and Failure Recovery (Section 4): investigating how AI-enabled automation can deliver self-healing and resilient networks, with the aim of ensuring service continuity while reducing operational complexity.

2. Architecture & Enablers

Achieving the vision of 6G requires rethinking network architecture, with AI and semantic intelligence integrated at the core of network design, operation, and service exposure. This section outlines the project's research directions and provides a survey of state-of-the-art architectural approaches and enabling technologies that support goal-driven, AI-native, and context-aware networking.

To start the discussion, Contribution 2.1 "Co-existence of Goal-Oriented and Legacy Users" focuses on enabling the coexistence of goal-oriented (semantic) and legacy (data-oriented) users in energy-constrained 6G networks. It introduces initial ideas for a unified framework that balances communication and computation resources, combining reconfigurable intelligent surfaces (RIS) and AI-driven radio resource management to adapt to semantic goals and network conditions.

Following this, Contribution 2.2 "6G Architecture meeting the requirements of the EU AI Act" discusses the exposure of AI/machine learning (ML) services within and beyond the network boundary, a key architectural enabler for future distributed intelligence. The section reviews current approaches to AI-as-a-Service, model sharing, and data access frameworks within 5G/6G research initiatives.

Continuing the analysis, Contribution 2.3 "From Fragmented Data to Semantic Reasoning in 5G Core" addresses the problem of interpreting heterogeneous core network data and enabling natural language interactions. It presents initial ideas for building semantic reasoning layers within the core and surveys the status of AI-driven analytics and semantic abstractions in large-scale network systems.

Finally, Contribution 2.4 "Combining RAN and Core Resources for Predictive Services" examines the architectural evolution of network functions (NFs) and the role of AI-augmented network data analytics function (NWDAF) for telemetry collection and dynamic NF profiling. The AI-augmented NF consists of a 3GPP standard NF such as NWDAF which has been enhanced with AI functionality for analyzing all the collected data. After briefly presenting planned studies, it reviews the state of the art in NF telemetry, NWDAF capabilities, and AI-based orchestration mechanisms.

2.1. Co-Existence of Goal-Oriented and Legacy Users

This contribution focuses on the co-existence of goal-oriented and legacy users, with an emphasis on energy-efficient operation in future 6G systems. The motivation arises from the growing heterogeneity of network services, where semantic communications introduce task-centric performance metrics that differ fundamentally from classical throughput-based approaches. To support the sustainable deployment of both paradigms, there is a clear need for: (i) a unified framework enabling effective co-existence and shared use of wireless and computational resources; (ii) strategies that explicitly target the minimization of power consumption; and (iii) mechanisms that explore and exploit the trade-offs between communication and computation.

2.1.1. Research Objectives

In this context, we plan to investigate two complementary directions: the use of RIS to improve energy efficiency by controlling the physical environment, and the development of AI-driven radio resource management that adapts to semantic goals and network conditions. This study will be anchored in the use-case "Highly efficient communication with light infrastructure," [6GARROW-D21] which reflects practical deployments where both resource constraints and service coexistence are critical. Our work aligns with the broader 6G vision of programmable, AI-native, and energy-conscious network infrastructures capable of supporting diverse, goal-driven services at scale.

2.1.2. Relevant State of the Art

The integration of goal-oriented (semantic) and legacy (data-oriented) services in 6G networks presents a significant research challenge, especially under energy constraints. Legacy systems

focus on throughput and bit-level reliability, while semantic systems prioritize task completion, meaning extraction, and contextual relevance. These diverging goals demand new paradigms for resource sharing, interference management, and power-aware optimization. We review the state of the art along several axes that underpin our contribution to Task 4.1.

Initial works have proposed coexistence strategies that rely on defining goal-centric performance metrics, such as goal-effectiveness and goal-cost [CSM+25]. These metrics enable dynamic resource allocation, ensuring semantic users fulfil their tasks while minimizing throughput degradation for legacy users. Such models exploit the robustness of semantic systems to interference, allowing shared spectrum access.

In [MFB+24], a cross-layer optimization framework jointly adjusts transmit power, computation resources, and inference delay, balancing semantic accuracy and legacy throughput. The recent study in [MF25] extends this approach by analysing feasibility regions where semantic goals can be met under joint power and latency constraints. It shows that goal-oriented transmissions can succeed with looser PHY-layer constraints, which opens room for flexible coexistence even in constrained environments.

Semantic communication inherently supports energy savings by enabling systems to “transmit less while achieving more.” By focusing on the meaning rather than the bit content, it is possible to reduce transmission load, required bandwidth, and decoding complexity.

In [HMC+24], semantic inference is shown to be robust even when using low-resolution RIS control (e.g., 1-bit phase quantization and partial surface activation), offering a path to low-power implementations. However, the joint optimization of RIS configurations for coexisting semantic and legacy services remains unexplored. No current study fully addresses how RIS could be dynamically controlled to balance power allocation and quality of service (QoS) across multiple user types.

In [MLG+23], a semantic-versus-bit power region is introduced that characterizes the minimum transmit power required to jointly satisfy semantic and legacy user requirements. Through analytical and simulation-based analysis, the authors demonstrate that semi-non orthogonal multiple access (semi-NOMA) consistently outperforms orthogonal multiple access (OMA) and NOMA, achieving both lower total power consumption and greater flexibility in handling rate and similarity trade-offs.

In parallel, [LC24] explores rate-splitting multiple access (RSMA) to support coexistence between semantic and traditional users. RSMA divides messages into common and private parts, which allows fine-grained interference management and accommodates heterogeneous user demands. While the study focuses primarily on rate region characterization, it shows that RSMA can achieve superior performance in terms of semantic similarity and bit rate trade-offs. However, energy minimization is not considered in the RSMA study, and RIS is not integrated.

The diversity of service requirements in 6G calls for AI-native resource management frameworks. While optimization-based methods (e.g., [MF25], [MFB+24]) have demonstrated promising results, they rely on full system knowledge and centralized decision-making. In contrast, AI-driven methods such as reinforcement learning or graph-based scheduling can adapt dynamically to semantic task success, user feedback, and environmental conditions.

Current literature lacks unified approaches that incorporate RIS control, semantic inference objectives, and real-time power efficiency, especially in multi-user scenarios. An open research direction is the integration of semantic goal awareness into learning-based access and power control policies, particularly in light infrastructure contexts where computing and sensing resources are limited.

2.2. 6G Architecture meeting the requirements of the EU AI Act

The integration of AI into 6G networks must go beyond performance to also ensure trustworthiness and compliance with regulation. The forthcoming EU AI Act [EUAIAct] will impose requirements on risk management, transparency, and human oversight, many of which apply directly to AI-driven network functions. This study will therefore explore the architectural enablers needed for 6G to meet

these requirements. The goal is to ensure that 6G architectures are not only AI-native and efficient, but also legally compliant and trustworthy, reinforcing Europe's leadership in secure and responsible AI for telecommunications.

2.2.1. Research Objectives

While the entirety of the EU AI Act will be applicable starting from 2 August 2027 as defined in Article 113 of the EU AI Act, certain parts enter into force earlier; for example, the section on prohibited practices is already active since 2 February 2025. In the present chapter, we specifically consider requirements related to General Purpose AI Models and specifically those of systemic risk as defined in Chapter V of the EU AI Act. The corresponding articles enter into force on 2 August 2025.

We furthermore build on the 6GARROW Use Case “**AI-as-a-Service for stakeholders & creation of tailored AI models according to End Users needs**” [6GARROW-D21]. In this context, new approaches are introduced related to AI/ML service provision intra-network and exposed externally such that consumers can access intra-network data and AI training capabilities to request tailored AI models as illustrated in Figure 1.

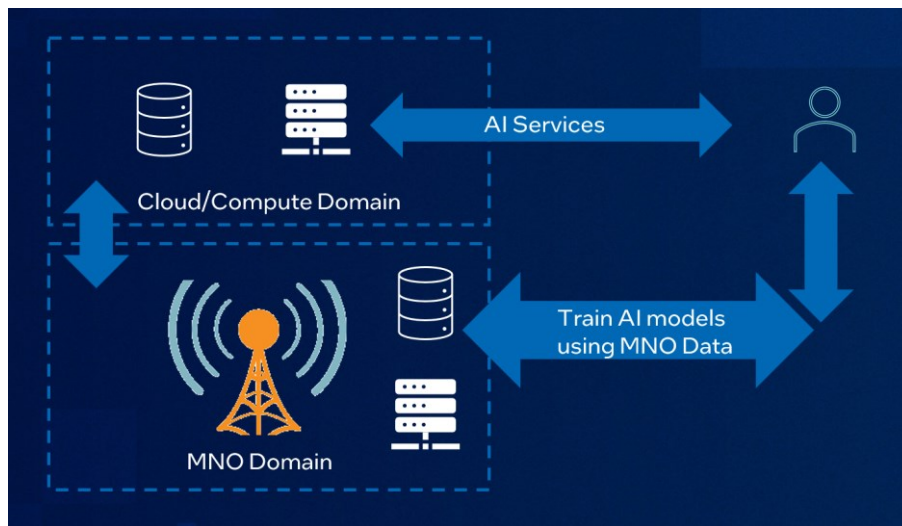


Figure 1: 6GARROW approach: MNO offers trained AI models exploiting MNO data, avoiding exposure of underlying raw training data.

We started to look at the requirements of the EU AI Act to offer “*AI/ML service provision intra-network and exposed externally*”. Especially, we examined what can be done in a Research Context vs a Commercial Usage context. Our intention is to propose an approach (including related architecture elements, features, etc.) that meets the requirements of the EU AI Act.

The EU AI Act includes the following recitals / articles on requirements specifically for application to research:

Recital (25): This Regulation should support innovation, should respect freedom of science, and should not undermine research and development activity. It is therefore necessary to **exclude from its scope the AI systems and models specifically developed and put them into service for the sole purpose of scientific research and development**. Moreover, it is necessary to ensure that this Regulation does not otherwise affect scientific research and development activity on AI systems or models prior to being placed on the market or put into service. As regards product-oriented research, testing and development activity regarding AI systems or models, the provisions of this Regulation should also not apply prior to those systems and models being put into service or placed on the market.

Article 2: Scope

[...]

6. This Regulation does not apply to AI systems or AI models, including their output, specifically developed and put into service for the sole purpose of scientific research and development.

...

The EU AI Act defines specifically “General Purpose AI Models with SYSTEMIC RISK”:

CHAPTER V - GENERAL-PURPOSE AI MODELS

SECTION 1: Classification rules

Article 51: Classification of general-purpose AI models as general-purpose AI models with systemic risk

1. A general-purpose AI model shall be classified as a general-purpose AI model with systemic risk if it meets any of the following conditions:

(a) it has high impact capabilities evaluated on the basis of appropriate technical tools and methodologies, including indicators and benchmarks;

(b) based on a decision of the Commission, ex officio or following a qualified alert from the scientific panel, it has capabilities or an impact equivalent to those set out in point (a) having regard to the criteria set out in Annex XIII.

2. A general-purpose AI model shall be presumed to have high impact capabilities pursuant to paragraph 1, point (a), **when the cumulative amount of computation used for its training measured in floating point operations is greater than 10^{25} .**

[...]

Those “General Purpose AI Models with SYSTEMIC RISK” need to meet stringent requirements:

SECTION 3: Obligations of providers of general-purpose AI models with systemic risk

Article 55: Obligations of providers of general-purpose AI models with systemic risk

1. In addition to the obligations listed in Articles 53 and 54, providers of general-purpose AI models with systemic risk shall:

(a) perform model evaluation in accordance with standardised protocols and tools reflecting the state of the art, including conducting and documenting adversarial testing of the model with a view to identifying and mitigating systemic risks;

(b) assess and mitigate possible systemic risks at Union level, including their sources, that may stem from the development, the placing on the market, or the use of general-purpose AI models with systemic risk;

(c) keep track of, document, and report, without undue delay, to the AI Office and, as appropriate, to national competent authorities, relevant information about serious incidents and possible corrective measures to address them;

(d) ensure an adequate level of cybersecurity protection for the general-purpose AI model with systemic risk and the physical infrastructure of the model.

[...]

We specifically address the following questions: What can be done if we have a General Purpose AI model of SYSTEMIC RISK which can be applied either i) in a scientific/research context or ii) in a commercial context? In a commercial context, a user/provider may choose to offer a “trimmed down version” to avoid additional requirements related to the “SYSTEMIC RISK” case.

An architectural approach will be considered which allows to select between a “full” and “trimmed down version” of an AI model such that a trade-off can be offered between i) performance of an AI Model and ii) inherent legislation requirements (which are considerably higher if the AI Model

qualifies as a Model of “systemic Risk”). The basic principle of the considered approach is illustrated in Figure 2.

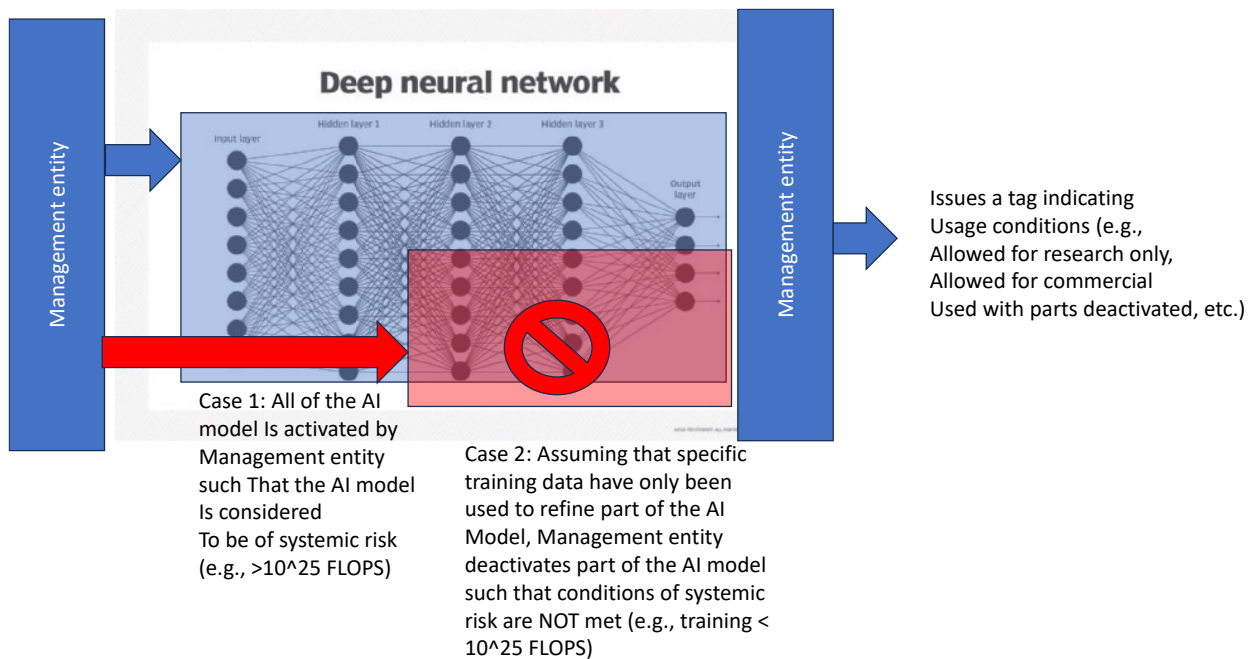


Figure 2: Offering the choice between a “full” and “trimmed down” version of an AI Model.

2.2.2. Relevant State of the Art

There is limited literature available related to the implementation of the EU AI Act [EUAIAct] and related architectural solutions. [MUECK+25] provides a generic overview of the AI Act and presents some architectural solutions to meet requirements for so-called high-risk AI systems. Li [QIAN22] and others have been discussing basic principles of cloud-native, compute-native and AI-native 6G Architectures. Corresponding solutions have mainly been focusing on extending the 3GPP service based architecture (SBA) by adding Compute and AI related entities, with a strong focus on intra-network usage. Furthermore, some principles of exposing related features have been introduced by Hexa-X on a high level [HEXAX+23]. However, related solutions are limited to intra-network processing to exploit mobile network operator (MNO) Data. We believe that the solutions developed in this framework by 6GARROW are highly novel and innovative and will significantly complement available state of the art literature.

2.3. From Fragmented Data to Semantic Reasoning in 5G Core

As we migrate to 6G networks, the demand to control enormous volumes of various data produced inside the 5G core (5GC) progressively challenges the mobile infrastructure. The 5GC is responsible for coordinating a wide range of functionalities such as user authentication, mobility management, session handling, and policy enforcement, all of which generate continuous streams of monitoring and control data. Driven by the increasing number of services, devices, and control layers, the growing complexity of network operations results in a flood of data coming from many different NFs. Often lacking semantic alignment and disjointed, this data is structured and unstructured and is rather difficult to understand or correlate across layers. Every NF runs alone, applying its own metrics and data formats, which prevents the network from expanding to provide a shared awareness of its general state.

Future networks expected to run under dynamic and large-scale multi-tenant conditions depend on intelligent and autonomous control, and this fragmentation greatly reduces the possibility for it. Without the ability to process and reason coherently over heterogeneous data, networks' robustness will continue depending on ineffective reactive mechanisms that are prone to failure under stress.

2.3.1. Research Objectives

What will be investigated are semantic and cognitive approaches to data integration and goal-oriented reasoning that allow the network to link, interpret, and act upon cross-domain knowledge in real time. This is essential because it enables the network to predict problems, maximize resource use, and guarantee a higher level of automation and resilience. These characteristics directly relate to the main objectives of the 6G vision, which include increasing energy efficiency, preserving service continuity, and enhancing user quality of experience (QoE).

The semantic state representation function (SSRF) contributes to solving these constraints, as detailed in deliverable D2.1 [6GARROW-D21], by means of a distributed cognitive system implemented inside the 5GC. Based on a multi-agent architecture, the SSRF can learn from unstructured and ordered data. This helps the system to link and deduce across several data sources, including those generated by many NFs, by means of several strategies. Unlike conventional intent-based management systems, which depend on centralized and essentially syntactic pipelines, SSRF provides semantic interpretation and goal-directed decision-making across many sectors.

A key difference compared to the approach by Wang et al. [YCT+25] is that their model focuses on a modular and layered architecture for intent translation and assurance, but it lacks an integrated reasoning capability over heterogeneous data. In contrast, SSRF enables distributed inference through cooperating agents and supports real-time interpretation of complex conditions using data from multiple, disjoint network functions. This difference helps the system to interpret user goals expressed in natural language and engage in suitable control logic across the network.

Semantic representations, large language models (LLMs), and autonomous agents help the network go from reactive behaviour to proactive cognition. Agent resilience and general quality of service can be raised by means of the analysis of observed conditions, integration of contextual knowledge, and application of autonomous decisions. This is a real step toward the 6G vision, which seeks intelligent and self-adaptable networks in addition to faster and more scalable ones. The SSRF enables reasoning over heterogeneous data and supports natural goal expression to help realize a fundamental aspect of this vision: networks that understand, predict, and act on high-level objectives with minimum human intervention.

2.3.2. Relevant State of the Art

Recent advancements in semantic and intent-based network management highlight efforts to reduce operational complexity and enhance automation within mobile networks as they evolve toward 6G. Much of the literature has concentrated on frameworks that map high-level user intents into actionable network configurations, often leveraging structured data pipelines and natural language interfaces. Examples include Habib et al.'s three-layer pipeline, which integrates LLMs, forecasting modules, and decision transformers for predictive orchestration [MPY+25], and the ALLSTaR architecture, which generates dynamic schedulers for RAN scenarios using LLMs [MMR+25]. Lam Dinh et al. further propose FEACI, a methodology for evaluating LLMs across multiple criteria, including inference efficiency, explainability, and format versatility [LSX+25]. Collectively, these approaches aim to introduce semantic abstraction and machine reasoning into network operations, improving adaptability and responsiveness.

However, current solutions exhibit notable limitations. Many approaches remain domain-specific, primarily focusing on the RAN, and rely on centralized processing pipelines that struggle to scale in large, heterogeneous, and dynamic 6G environments. Crucially, they lack mechanisms for reasoning over unstructured or cross-domain knowledge, and they rarely provide real-time coordination across multiple disjoint data sources. Additionally, no existing framework integrates a native multi-agent architecture capable of distributed inference directly within the 5GC. This gap underscores the need for a semantic and distributed reasoning layer that can guide network control actions autonomously, interpret user goals, and correlate heterogeneous observability signals in real time.

The semantic state representation function (SSRF), introduced in deliverable D2.1 [6GARROW-D21], addresses these gaps by embedding a multi-agent cognitive layer directly into the 5GC. SSRF enables reasoning over both structured and unstructured data from diverse NFs, facilitating coordinated decision-making across multiple domains, semantic correlation of network states, and real-time interpretation of goals expressed in natural language. Unlike prior systems, SSRF supports proactive, agent-driven behaviour grounded in a shared semantic representation, providing a concrete architectural approach toward the 6G vision of intelligent, self-optimizing, and context-aware networks capable of understanding and acting on high-level human-defined objectives.

2.4. Combining RAN and Core Resources for Predictive Services

To ensure the commercial success of 6G, it is essential to provide new services and offerings to clients an added value that would enable their business growth. A weakness of 5G new radio (NR) has been that commercial users often did not experience a notable difference of the communication services that were offered when compared to 5G, which did not add any new value for which they were willing to pay. Furthermore, capital expenditure (CAPEX) and operational expenditure (OPEX) related metrics must be improved to support sustainability objectives and increase commercial viability. Applying an “*AI-native*” approach to the 6G architecture design will be key to reaching these objectives by ensuring low-latency and ubiquitous access to intelligence. This would allow: i) The introduction of new functionalities and services to end users; and ii) The reduction of the CAPEX and OPEX to deploy and operate the 6G infrastructure, respectively.

2.4.1. Research Objectives

Network operators collect large amounts of data while operating their mobile network. The data is collected by operations, administration, and maintenance (OAM) functions and by NWDAF of the core network. NWDAF was introduced to collect data from core network NFs to provide immediate, actionable analytics to optimize network performance and enhance the user experience. NWDAF enables AI-driven automation and improved network management by providing analytics services to other core network NFs as well as acting as a bridge between OAM data services that have focus on longer term operations. It supports distributed architecture suitable for edge computing as defined in [TS23.288] and can be used for federated learning. AI functionality needs to be implemented beyond the scope of today's NWDAF for the purpose of both network automation and end user applications. For instance, the RAN could provide RAN radio resource consumption history of an application data to NWDAF which can then train AI models to deal with the given application.

Since the NWDAF together with data collection functions of OAM such as management analytics function (MDAF) have the possibility to create projections of the future usage of resources in the different parts of the mobile network, 6G services will have the possibility to extend the services offered so far to predict future demands. However, some important and time sensitive radio related data, such as channel state information (CSI) and sounding reference signal (SRS) measurements, are still collected by the base stations (BSs) of the RAN. Such data ages require very short, closed control loops limiting their usability beyond radio layer optimization. However, CSI and SRS together with other lower layer data can contribute to services like positioning and spectrum sharing among others. Therefore, it is very important to unify the RAN and CN data collection, such that information can be consistently collected, processed and enriched to be consumed in different time scales and aggregation levels by the performance critical functions as well as a value-added auxiliary information depending on the needs of the data consumer. Value added information would be an outcome of combining and processing the raw data to meet the needs of the consumer (e.g., adding labels and semantics on data). The data consumer can be a network internal entity (e.g. an AI-agent) or an external service provider (i.e., Digital Twin or another network operator sharing same frequency bands). Distributed NWDAF would be the basis for new 6G NWDAF that combines RAN, CN and edge analytics for various end-to-end optimizations. In most real-life deployments RAN and CN is required to be kept as logically separate administrative domains because of business reasons (e.g.,

they may have different owners or providers) and technical requirements including the need to operate in different time scales and security context. A possible approach to unified data collection across RAN – CN boundary while respecting the RAN – CN separation is to base 6G NWDAF on federated learning approach that is a decentralized approach where the data stays on their local domains, and only AI/ML-model updates are exchanged between RAN and CN. Within a domain distributed learning approach can be applied to process large dataset distributed across multiple compute nodes to speed up training.

The value of the collected data that is exposed for network external services and applications depends on the accuracy/reliability, timeliness and format of the exposed data. These attributes are likely to vary from one external service to another (e.g., spectrum usage vs. digital twin of autonomous vehicles). Access to collected data is provided through “Exposure Layer” that is a generalization of the network exposure function (NEF) exhibiting exposure application programming interfaces (APIs) for data collection and analytics. When an external service provider requests a specific service-related information through the APIs of the “Exposure Layer”, the requests should contain information about the accuracy and scope (e.g., location, time or type of user) of the service data as well as the expected format. Alternatively, the exposure API can be intent-based, leaving the data request configuration details and validity assessment to the exposure layer. Exposure layer validates the requests and authorizes them after checking their compliance against the existing policies such as GDPR, EU Data Act, including privacy considerations and operator own policies.

The exposure Layer would be interacting with an AI orchestrator that has overall visibility of available resources in different parts of the system such as Radio Access, Transport, Edge Computing and Core network functions. The AI orchestrator or oracle would be able to search and discover system resources and interact with distributed AI agents to reconfigure them as required by the request from the exposure layer. The Figure 3 below shows the subsystems and resources with own distributed controllers and AI agents that will interact with the AI oracle to re-configure the resources to support different use case and service requirements. Resources from the pools of infrastructure are discovered and managed via LLDP and NETCONF protocols. RESTFULL-interfaces are used to configure AI agents and to communicate with the controllers.

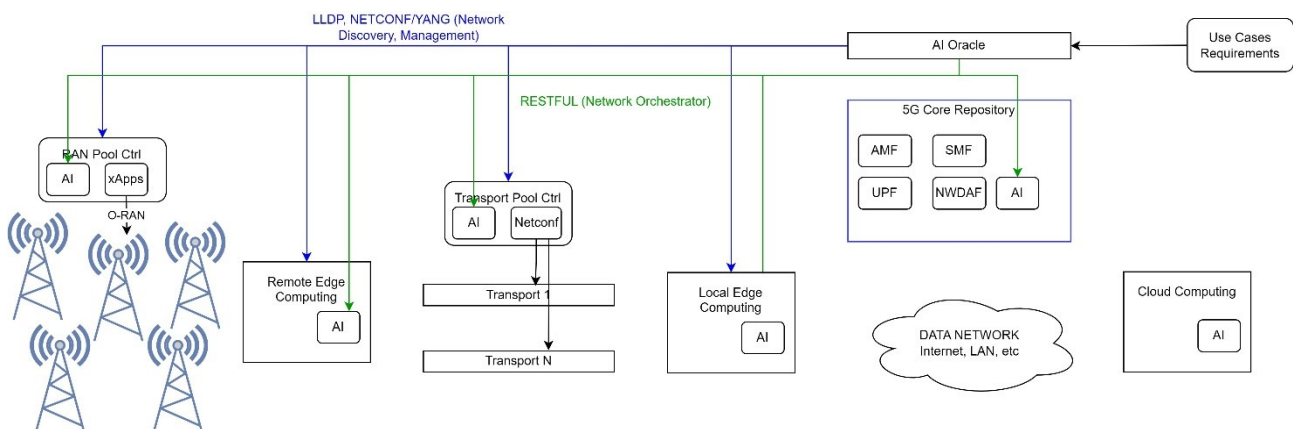


Figure 3. Native AI designed mobile infrastructure with distributed subsystem AI agents.

2.4.2. Relevant State of the Art

5G Releases 18 defines a distributed architecture for NWDAF by introducing Analytics Logical Function and Model Training Logical Function logical components. New supporting functions, such as the data collection coordination function (DCCF) and analytics data repository function (ADRF), were added to handle scalability and efficiency of data collection. NWDAF is thus becoming a central entity for deploying AI-models and agents [TS23.288]. For 6G, NWDAF is expected to evolve into a fully distributed "cognitive fabric" where intelligence is embedded across the network—from the core to the edge.

5G NEF [TS29.522] acts as a gateway, translating internal network data and functionality into standardized, open, CAPIF, based APIs. NEF allows several types of applications to make use of the mobile network to carry out specific services such as Network Events (e.g., UE location, UE roaming status) exposure, QoS and application traffic influence within the network [EXP25]. As the number and diversity of services is expected to grow in 6G, NEF needs to evolve to a cross-layer Exposure Layer that encompasses RAN, CN, edge and possibly UEs.

As a concrete example of data that has high potential value is sensing data collected from the RAN and matched against target object fingerprints for the purpose of object recognition. Integrated sensing and communication (ISAC), also known as joint communications and sensing (JCAS), has been identified one of the key usage scenarios for 6G [TS22.137]. Hexa-X has been studying the impact of JCAS for different device types. ISAC/JCS is seen as a wireless communication system used to also provide the sensing capabilities that is applicable across different device classes and based on specific use-cases/scenarios. Sensing data can be based on CSI that describes instantaneous properties of the wireless channel and is the basis of any beamforming used in multiple input multiple output (MIMO) based solutions [KSS+24]. Beyond its typical use for enabling optimal subcarrier resource allocation for OFDMA, CSI augmented with other JCAS attributes (such as location, doppler) has been studied as a complementary component for sensing for multiple sensing tasks. For example, in XR applications, CSI could be used to get a (near) real-time understanding of the environment that a user is traversing or in industrial scenarios where robots can detect a human nearby [NGJCAS24].

3. AI Native 6G Resource usage

One of the defining characteristics of 6G is the tight integration of AI into the radio and physical layers, enabling dynamic adaptation to traffic, mobility, and channel conditions. This section outlines the planned avenues of investigation on short-term, localized AI algorithms for resource optimization. It also presents a review of the state of the art, focusing on current methods for AI-based resource allocation, physical-layer intelligence, and advanced channel modeling.

Contribution 3.1, “Slice-Aware AI-Driven Energy Saving,” proposes to study multi-scale AI for QoS-aware RU ASM management and network slicing state management, by considering both long-term traffic trends and short-term mobility dynamics. The planned work will explore both macro-level adaptations (e.g., slice configuration for RUs) and fine-grained decisions (e.g., ASM of RUs). The state of the art in AI-driven RRM across different scales is then examined, with attention to latency, scalability, and mobility support.

Contribution 3.2, “AI/ML Physical Layer Techniques for 6G Air Interface,” focuses on applying AI/ML at the physical layer to improve energy efficiency and spectral performance. The research will investigate the use of AI for CSI prediction and beam management, leveraging both simulated and field data. This is complemented by a review of current physical-layer AI implementations, testbeds, and standardization efforts that reflect the maturity and feasibility of such approaches.

Contribution 3.3, “Cell-specific Digital Twin,” aims to develop site-specific digital twin using Sionna. The proposed work involves realistic channel generation, conditioned on factors like user location and environmental parameters. The state of the art in channel modeling with AI—including generative approaches, spatial consistency, and data-driven simulation—is reviewed to position the proposed work within ongoing research trends.

3.1. Slice-Aware AI-Driven Energy Saving

Reducing energy consumption is a critical goal for future RANs, offering significant environmental and economic benefits. With base stations currently contributing to approximately 80% of total RAN energy usage, innovative power-saving strategies are essential for sustainable network evolution.

The architectural shift towards the O-RAN, with its disaggregated components and programmable RIC, creates an unprecedented opportunity for intelligent, AI-driven optimization. This paradigm,

combined with network slicing, allows operators to deploy multiple virtual networks over a shared infrastructure, each customized with specific QoS guarantees for diverse applications like eMBB and URLLC.

3.1.1. Research Objectives

The main research challenge lies at the intersection of these domains: how to maximize energy savings through aggressive sleep management without compromising the strict SLAs of co-existing network slices. This conflict is most acute for URLLC services, where the wake-up latency from deep sleep states would introduce unacceptable delays, violating their core performance guarantees.

As shown in Figure 4, this research aims to develop a novel, slice-aware, AI-driven framework for ASM, designed for deployment within the hierarchical O-RAN RIC structure. The framework employs a two-tiered AI control loop operating across different timescales:

1. **Strategic Management (Non-RT):** At the Non-RT timescale, an rApp will perform strategic slice topology and resource orchestration. Analyzing historical traffic patterns and service demands, it will intelligently decide which network slices are activated on which RUs and pre-allocate coarse-grained resources. A key objective is to consolidate URLLC services onto a minimal set of dedicated RUs, thereby limiting the network footprint that must remain "always on" and instantly available.
2. **Tactical Control (Near-RT):** At the near-real-time timescale, an xApp will execute dynamic ASM policies under the strategic constraints set by the rApp. By monitoring live, per-slice traffic, it will intelligently transition RUs between active and sleep states. Critically, any RU hosting an active URLLC slice will be prevented from entering deep sleep modes to guarantee low-latency performance. Conversely, RUs serving only eMBB or mMTC slices will be subject to aggressive deep sleep policies to maximize energy conservation.

These interdependent AI models will be trained and rigorously validated within an NVIDIA Sionna-assisted digital twin environment. This high-fidelity simulation will model a multi-slice O-RAN deployment, ensuring the framework can robustly balance the twin goals of energy efficiency and guaranteed per-slice QoS before real-world implementation.

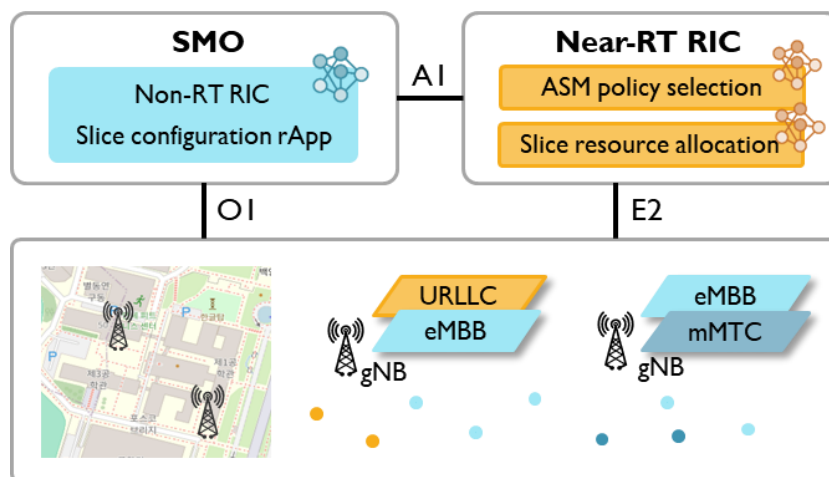


Figure 4. Slice-Aware AI-Driven ASM and resource allocation

3.1.2. Relevant State of the Art

The escalating energy consumption of the RAN, driven by the explosive growth in high-demand services like URLLC and eMBB, has become a primary concern for network operators. Consequently, developing energy-efficient solutions that uphold stringent QoS has emerged as a critical field of research. The flexible, disaggregated architecture of O-RAN, in particular, has opened new frontiers for intelligent power management.

A foundational step in this area is the accurate modeling of power consumption. The work in [AHME25] provides a crucial analysis by measuring the power draw of disaggregated O-RAN components across different functional splits (specifically Split 8 and Split 7.2b). Their findings underscore the significant potential of dynamic sleep management—strategically switching components on and off—as a primary mechanism for energy savings. Building on this, managing the sleep modes of disaggregated RAN components has become a major research focus, as it is one of the most effective techniques for reducing operational expenditures.

As the complexity of network optimization grows, researchers have turned to advanced computational methods. For instance, [GAO25] proposes a joint load adjustment and sleep management scheme for virtualized RANs. Their approach employs a quantum genetic algorithm—noted for its superior efficiency over traditional GA—to solve a multi-objective problem: maximizing computing power efficiency while minimizing the PUE of computing nodes.

More recently, the integration of AI, particularly DRL, has shown immense promise for managing dynamic network environments. DRL is exceptionally well-suited for scenarios with unpredictable traffic loads and fluctuating resource availability. In [GAN25], a DRL-based strategy is proposed for joint sleep control and energy sharing in ultra-dense networks composed of SBS. The authors demonstrate how a PPO algorithm can effectively navigate the large and complex action space inherent in controlling a multitude of SBSs, adapting its policies in real-time. To further enhance system intelligence, the GreenRAN framework introduced in [YOU25] executes a two-step, channel-aware power-saving process. It combines a Channel-Aware RU ON/OFF Controller as an rApp with a CU-DU Placement Module as an xApp, creating a holistic system that minimizes power consumption while providing QoS guarantees.

Despite these significant advances, a critical gap remains in the literature. Much of the existing research either pursues network-wide energy efficiency without considering the distinct and often conflicting SLAs of different network slices, or it focuses on slice-level resource management for QoS without fully exploiting aggressive sleep strategies as a primary energy-saving tool. A key limitation is the lack of an integrated framework that is explicitly slice-aware for the purpose of deep sleep management. We identify a major open challenge in designing a hierarchical AI framework that jointly performs strategic slice configuration and tactical, slice-aware sleep management. Our work aims to address this gap directly by creating an AI-native solution that can expertly navigate the trade-off between guaranteeing URLLC slice availability and maximizing the energy savings from RUs dedicated to less latency-sensitive services.

3.2. AI/ML Physical Layer Techniques for 6G Air Interface

This section introduces how AI/ML is being integrated into the physical layer of next-generation wireless systems and how to design and validate AI/ML models and algorithms for the physical layer of 5G-Advanced and 6G. The discussion focuses on beam management and CSI prediction, which will be implemented in the ETRI testbed as shown in Figure 5. These are essential for improving energy efficiency and resource utilization. The relevant state of the art shifts from traditional model-based methods to data-driven solutions. Recent advances apply deep neural networks [AHMA24-KARU24], generative AI [LIAN24, ZZLI25], and reinforcement learning [JIAN24-XLIU25] to massive MIMO, mmWave/THz operation, ISAC waveform design, and real-time beam tracking for mobility. Subsequent subsections explain the research objectives and offer comparative summaries (Table 1 and Table 2) to assist in selecting and integrating models for beam management and CSI prediction.

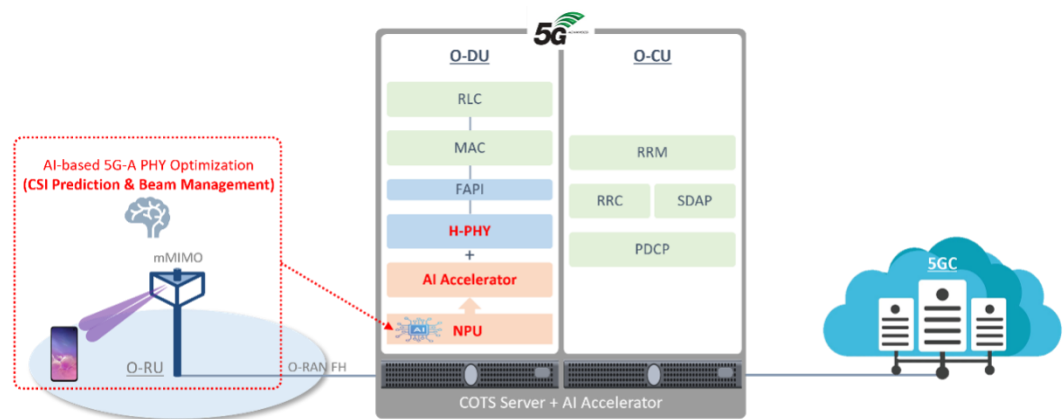


Figure 5.AI/ML physical layer techniques in ETRI testbed

3.2.1. Research Objectives

The objective of this research is to design, implement, and validate AI/ML models for the physical layer of 5G-Advanced and 6G wireless communication systems. These models will target the O-RAN 7.x split and will execute all AI inference on the O-DU, maintaining O-RU behavior to ensure clean and interoperable integration with open standards. The models will be optimized for real-time deployment in the ETRI O-DU testbed and will be validated in the 3.5 GHz/100 MHz band under MIMO operation without fixing a specific antenna count. The study will focus on two key technical areas—beam management and CSI prediction—to reduce search overhead and beam-switch latency, sustain scheduler visibility with fewer pilots and feedback, and improve energy and resource efficiency. The implementation will be integrated into the ETRI testbed and evaluated using two primary indicators—energy consumption and inference—to demonstrate practical effectiveness, interoperability, and scalability in real-world deployments.

3.2.2. Relevant State of the Art

The integration of AI/ML into the physical layer of wireless communication systems has progressed rapidly, playing a pivotal role in the evolution of 5G-Advanced and laying the foundation for 6G networks. Departing from traditional analytical approaches, AI/ML enables data-driven solutions to address complex physical layer challenges such as channel estimation and prediction, signal detection, beam management, CSI compression and feedback, interference mitigation, and dynamic resource allocation. Recent advances demonstrate the application of neural networks, generative AI, and RL to emerging physical-layer challenges such as high-dimensional signal processing in massive MIMO, joint waveform design for ISAC, and real-time beam tracking to support mobility.

Beam management is one of the most critical challenges in 5G-Advanced and 6G wireless communications, especially in the mmWave and THz frequency bands. The use of massive antenna arrays and highly directional beams necessitates intelligent mechanisms for beam selection, tracking, and prediction to ensure reliable communication in dynamic and complex environments. Conventional beam management techniques often suffer from high signaling overhead, increased latency, and limited adaptability to rapidly varying channel conditions, motivating the adoption of AI/ML-based solutions. Table 1 provides a comparative overview of recent AI/ML approaches to beam management, summarizing their learning strategies, model architectures, and key innovations.

Table 1. Comparative Analysis of AI/ML Beam Management Methods

Learning Method	Models	Key features	Ref.
Supervised Deep Learning	CNN	98% beam prediction accuracy in LOS scenarios using DeepSense 6G dataset	[AHMA24]

Supervised Deep Learning	LSTM	Reduces beam tracking latency by 85% through temporal dependency analysis	[JAYA24]
Supervised Deep Learning	CNN + Vision Processing	End-to-end vision-aided beamforming with 98.96% top-5 accuracy, geometric adjustment for camera distortion	[KARU24]
Deep Reinforcement Learning	DQN	Sensing-echo-based multi-user beam assignment, throughput improvement up to 407% compared to conventional beam sweeping	[WANG24]
Multimodal Learning	Transformer + RL	Integrates LiDAR, GPS, and camera data for robust beam prediction in dynamic environments	[MARE24]
Generative Learning	GAN	Infers optimal beam/AP from a small subset of beams; data-efficient inference for mmWave D-MIMO systems	[KART24]
Federated Learning	Generative FL	Addresses label/modality imbalance via adaptive zero-shot multi-modal data generation & hybrid training paradigm	[LIAN24]
Hybrid Deep Learning	CNN-LSTM	Spatial-temporal beam prediction (SBP/TBP) models for 5G-Advanced, achieving 98% accuracy in system-level simulations. Complies with 3GPP Release 18/19 standards.	[JAYA24]

Recent AI/ML-based approaches have significantly advanced beam management in 5G-Advanced and 6G wireless systems by addressing the limitations of traditional methods. One such approach leverages convolutional neural networks (CNNs) to predict the optimal beam in line-of-sight scenarios, achieving up to 98% accuracy and enhancing reliability in massive MIMO deployments [AHMA24]. Another method employs long short-term memory (LSTM) networks to capture temporal patterns in the wireless channel, resulting in an 85% reduction in beam tracking latency and enabling faster adaptation to dynamic environments [JAYA24].

Vision-aided beamforming has also gained attention, where CNNs are combined with vision processing to enable end-to-end beam selection [KARU24]. These solutions achieve a top-5 prediction accuracy of 98.96% and apply geometric corrections for camera distortion, improving the robustness of vision-based beam management. Deep reinforcement learning (DRL) techniques—particularly deep Q-networks (DQNs)—have been applied to multi-user beam assignment using sensing echoes, yielding up to a 407% increase in throughput compared to conventional beam sweeping, especially in dynamic, multi-user scenarios [WANG24].

Multimodal learning strategies further enhance performance by integrating data from various sensors such as LiDAR, GPS, and cameras, using transformer-based models combined with RL [MARE24]. This multimodal fusion enables accurate beam prediction in highly dynamic environments where reliance on a single sensing modality may be insufficient. Generative adversarial networks (GANs) have been employed to infer optimal beams or access points from a limited set of candidates, supporting data-efficient inference in mmWave distributed MIMO systems where exhaustive search is impractical [KART24].

Federated learning approaches tackle issues such as label imbalance and heterogeneous sensing modalities by using generative models that enable adaptive zero-shot multi-modal data generation and hybrid training strategies. These techniques support robust beam management even under conditions of incomplete or imbalanced data across devices [LIAN24]. Finally, hybrid models combining CNN and LSTM architectures have been developed for spatio-temporal beam prediction in 5G-Advanced systems. These models not only achieve 98 % accuracy in system-level simulations but also comply with 3GPP Release 18/19 specifications, ensuring both high performance and standards compatibility [JAYA24].

Accurate CSI prediction is essential for optimizing the performance of advanced wireless systems, including 5G-Advanced and future 6G networks. In recent years, significant progress has been made in AI-based CSI prediction, with deep learning and RL techniques employed to tackle challenges such as channel aging, feedback overhead, and the complexity of dynamic wireless environments. Table 2 provides a comparative overview of recent AI/ML-based CSI prediction approaches, summarizing their learning paradigms, model architectures, and key innovations.

Table 2. Comparative Analysis of AI/ML CSI Prediction Methods

Learning Method	Models	Key features	Ref.
Supervised Deep Learning	CNN	Federated/distributed CNNs for RIS-assisted MIMO; strong spatial feature extraction	[KORA25]
Supervised Deep Learning	LSTM	Sequence-to-sequence LSTM for predicting time-varying CSI in massive MIMO; improved robustness to channel aging and feedback delay	[WANG25]
Supervised Deep Learning	Transformer	Multi-head self-attention with channel masking; robust to missing pilot data	[JKIM25]
Deep Reinforcement Learning	DQN/DDQN	Joint CSI prediction and resource optimization; handles dynamic, multi-user scenarios	[JIAN24]
Deep Reinforcement Learning	DDPG	Continuous action space; used in advanced resource allocation and prediction tasks	[XLIU25]
Large Language Model	LLM, LLM4CP	Pretrained sequence modelling; modality alignment for CSI features; robust to noise, few-shot capable	[ZZLI25]
Hybrid Deep Learning	CNN-LSTM	Encoder-decoder CNN with LSTM and skip connections; robust to Doppler and channel model mismatch; enables pilot reduction in uplink OFDM for 6G NTN	[FILI25]

Among the various AI/ML techniques explored for CSI prediction, several model architectures have demonstrated particular strengths in addressing key challenges across different deployment scenarios. CNNs are widely adopted due to their strong spatial feature extraction capabilities, especially when implemented in federated or distributed architectures for RIS-assisted MIMO systems [KORA25, WANG25]. LSTM networks, particularly in sequence-to-sequence configurations, are effective at modelling time-varying CSI in massive MIMO systems by capturing temporal dependencies and mitigating the effects of channel aging and feedback delays [ZZLI25]. Transformer models, leveraging multi-head self-attention and channel masking, provide robustness to missing pilot data and excel at capturing long-range dependencies in wireless communication tasks [KORA25, WANG25].

DRL approaches, such as DQN and double DQN (DDQN), integrate CSI prediction with resource optimization by adapting to dynamic, multi-user environments through real-time feedback [JIAN24], while deep deterministic policy gradient (DDPG) extends these capabilities to continuous action spaces, supporting fine-grained resource allocation in more complex settings [XLIU25]. Recently, LLMs, such as LLM-empowered channel prediction method (LLM4CP), have been adapted to CSI prediction by leveraging pretrained sequence modelling and modality alignment, offering robustness to noise and strong generalization capabilities from limited data via few-shot learning [ZZLI25, FILI25]. Finally, hybrid CNN-LSTM architectures combine the spatial learning strengths of CNNs with the temporal modelling capabilities of LSTMs in encoder-decoder frameworks. These models often incorporate skip connections to improve robustness against Doppler shifts and channel mismatches, which is particularly beneficial for reducing pilot overhead in uplink OFDM transmissions for 6G non-terrestrial networks [FILI25].

Despite recent advances, several critical challenges remain in applying AI/ML to beam management and CSI prediction for 5G-Advanced and 6G systems. Many existing approaches demonstrate high performance in controlled environments but fall short in practical deployment scenarios characterized by user mobility, dynamic channel conditions, and multi-user interference. Moreover, most state-of-the-art models are computationally intensive, limiting their applicability in real-time systems, especially when deployed on resource-constrained edge devices. These limitations highlight the importance of identifying AI/ML models that are not only accurate but also lightweight, scalable, and feasible for real-world implementation. In particular, there is a critical need for models that balance high performance with low deployment complexity—models that can generalize across heterogeneous environments, require minimal labeled data, and operate with low latency and energy consumption. Their practical importance is further emphasized by the goal of deploying them on the ETRI-developed testbed, where interoperability, modularity, and standards alignment are essential.

To address these gaps, the proposed work focuses on developing efficient, high-performing AI/ML models for beam management and CSI prediction that are tailored for real-time operation. The models are designed to enhance adaptability, robustness, and data efficiency in varying wireless conditions. Furthermore, the approach emphasizes lightweight and scalable deployment strategies to enable distributed inference with minimal overhead. By aligning the architecture and implementation roadmap with ongoing 3GPP and O-RAN standardization efforts, this research aims to deliver AI-native solutions that are readily deployable on the ETRI testbed and applicable to future wireless networks.

3.3. Cell-specific Digital Twin

This work introduces the concept of Y-Twin, which is a research-oriented digital twin designed to replicate the behavior of a cellular RAN. Unlike deployment or operations tools, Y-Twin emphasizes realism, controllability, and extensibility, making it an ideal testbed for AI-native RAN research. It integrates ray-tracing based channel modeling, mobility and traffic simulation with 5QI service differentiation, as well as PHY/MAC-level observability, all within a flexible and reproducible framework. Y-Twin ultimately aims to generate high-quality data and provide controlled evaluation environments, tackling key challenges of data scarcity and benchmarking bottlenecks in wireless AI research.

3.3.1. Research Objectives

The overarching goal of our work is to design and implement Y-Twin, a digital twin system that faithfully replicates the behavior of a cellular RAN within a specific urban setting that, without loss of generality, is assumed to be in the Sinchon district of Seoul, South Korea. Y-Twin is not intended as a deployment or operations tool; instead, it is built as a research-centric framework that prioritizes realism, controllability, and extensibility. This makes it particularly valuable for advancing the AI-native RAN paradigm, where learning-based methods depend on high-quality training data and systematic evaluation environments.

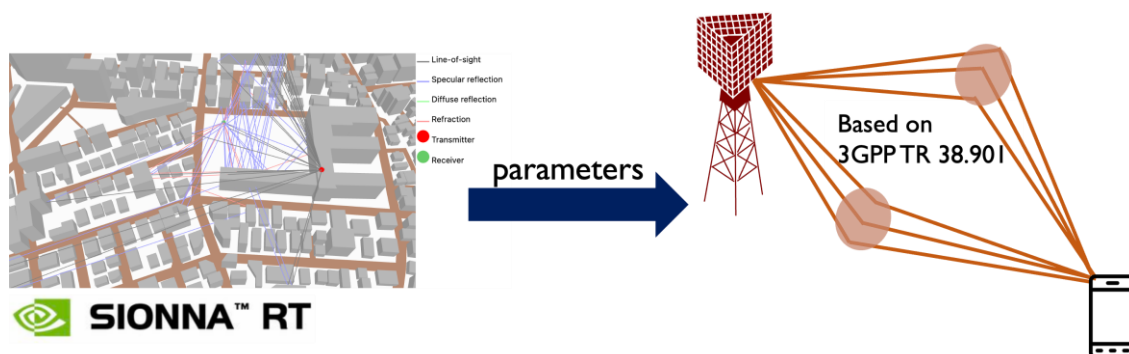


Figure 6. Channel modelling from Sionna RT results based on 3GPP TR 38.901.

At the heart of Y-Twin is a channel generation pipeline that directly incorporates the geometric blockage environment of the Sinchon district. This is achieved by adopting Sionna RT, NVIDIA's differentiable ray-tracing engine, which produces multipath channel parameters such as angles of departure/arrival, path delays, and path powers derived from physical geometry. To ensure compatibility with industry standards and reproducibility, these ray-tracing outputs are calibrated against the 3GPP TR 38.901 channel model [TR38901], which defines large-scale fading statistics across multiple deployment scenarios (e.g., urban microcell, urban macrocell). A simple schematic representation of this can be found in Figure 6. This hybrid approach enables the possibility to bridge the gap between deterministic site-specific modeling and standardized stochastic models.

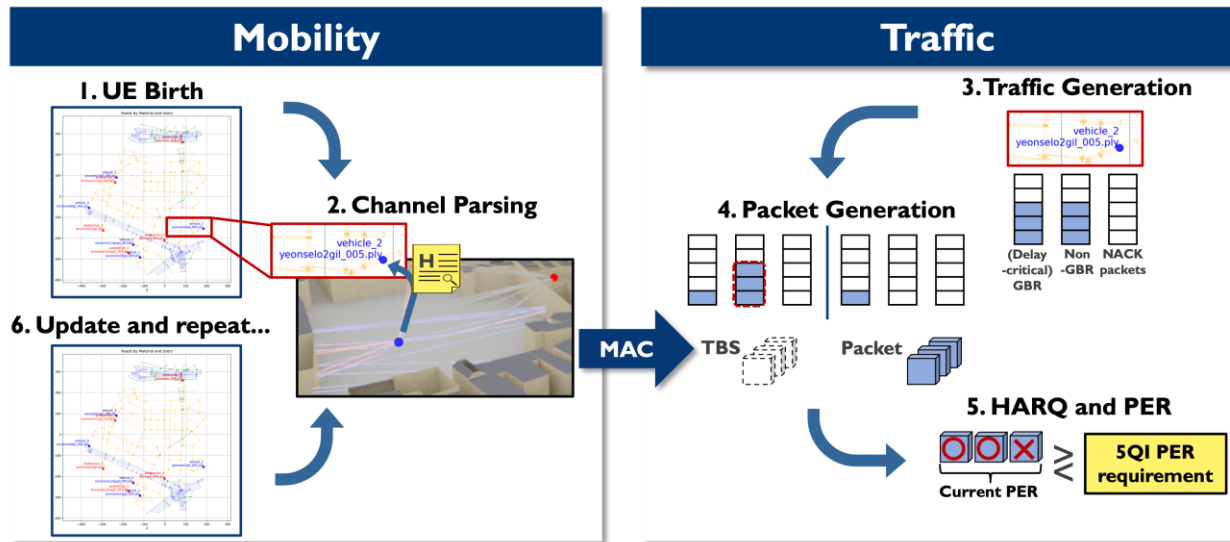


Figure 7. Overall process of mobility and traffic interaction.

Y-Twin explicitly models pedestrian and vehicular mobility based on realistic patterns found in the Sinchon area of Seoul. More importantly, traffic demand is not modeled as generic throughput curves but, instead, it is directly parameterized using the 5QI framework from 3GPP TS 23.501 [TS23501]. This allows each user to be assigned given service-level objectives, such as latency bounds, error rate constraints, and throughput requirements. By embedding 5QI into the digital twin, it is possible to reproduce not only physical channel conditions, but also the service differentiation that governs the way in which contemporary mobile networks allocate resources. Figure 7 illustrates the overall process in which the mobility and traffic management of a new UE that requests mobile service in the target area are jointly executed.

The digital twin presented in this section integrates PHY-layer functions such as MU-MIMO beamforming, channel estimation, and link adaptation, as well as MAC-layer functionalities such as scheduling and resource allocation. Unlike site deployment twins, Y-Twin goes one step further by producing per-user decoded bits rather than just signal strength or coverage metrics. This enables us to measure communication performance directly in terms of data integrity, error rates, and latency. Importantly, the scope is intentionally focused, and for the moment, higher-level RAN components (e.g., RIC or core network simulators) are not included in this analysis. By concentrating on PHY and selected MAC aspects, a clean, controlled environment for evaluating algorithms without introducing additional confounding factors is created.

Y-Twin is designed with the guiding principle of enabling flexible reusability. This way, simulation scenarios can be replayed under varying conditions, and modules such as schedulers, beamformers, or channel estimators can easily substitute or extend. This feature distinguishes Y-Twin from operational twins, which are often constrained by the fidelity of real-time inputs or by vendor-specific toolchains. The framework introduced in this section enables researchers to iterate rapidly on new ideas, test them under consistent conditions, and reproduce results for peer validation.

The critical motivation for Y-Twin is to serve as a data generator and evaluator for AI-native RAN. Live networks rarely provide the kind of annotated, privacy-safe, large-scale datasets required for training ML models. Y-Twin addresses this by producing realistic synthetic data that can mimic real-world distributions, while avoiding the exposure of sensitive user information. At the same time, the Y-Twin concept provides an evaluation sandbox, where new algorithms (e.g., AI-aided channel estimation, reinforcement learning-based scheduling) can be tested against repeatable scenarios. In this way, Y-Twin directly tackles two of the most critical bottlenecks in AI-native RAN:

- **Data scarcity**, due to the difficulty of obtaining relevant RAN data in large scale for training AI/ML models;
- **Evaluation bottlenecks**, due to the the lack of controlled environments for benchmarking new radio resource management methods.

Taken together, these objectives position Y-Twin as a strategic enabler for advancing data-driven wireless research. By emphasizing realistic user behavior, service-aware traffic, and controllable RAN emulation, the Y-Twin concept provides exactly the environment needed to bridge the gap between theoretical AI/ML models and deployable AI-native architectures.

3.3.2. Relevant State of the Art

The concept of DTs in wireless networks has gained momentum over the past few years, driven by both industry and academia, and is surveyed in [AFP+22] in a broad context. However, existing platforms tend to be specialized for either **planning and deployment** or **operations and assurance**, rather than for **research-driven AI experimentation**.

State of the Art regarding Industry platforms.

- **NVIDIA aerial omniverse digital twin (AODT)**: This platform targets **city-scale network planning and optimization**, integrated with Omniverse for 3D scene simulation. It excels at simulating entire metropolitan deployments, but its focus is primarily on **operator workflows** (coverage planning, capacity analysis) rather than decoded-bit observability.
- **Keysight network DF**: This is a platform geared toward **operator assurance and anomaly detection**, often used alongside O-RAN frameworks. It leverages measurement-driven validation and AI tools for ensuring service reliability. However, its primary outputs are **network KPIs**, not the detailed per-user data streams needed for training ML/RL algorithms.
- **Nokia and Ericsson site twins**: These platforms are focused largely on **cell-site deployments, indoor planning, and coverage optimization**. These DTs provide valuable insight into rollouts and material/structural impacts, but they stop short of delivering **communication-layer outputs** (e.g., decoded user bits) or **service-aware traffic differentiation**.
- **O-RAN digital twin initiatives**: Position digital twins [PBD+23] as enablers for AI training and evaluation in AI-native RANs. However, much of this work remains **conceptual** or **framework-level**, lacking concrete implementations with PHY/MAC fidelity.

Academic and open-source ecosystems.

- **Sionna and Sionna RT**: Provide differentiable PHY modeling and ray tracing, enabling detailed link-level simulations. While powerful, Sionna is primarily a library component, whose integratoin into a **full-stack digital twin** requires additional effort.
- **ns-3, 5G-LENA, and Simu5G**: These are widely used event-driven simulators that support system-level RAN modeling. However, their channel models are often represented in the form of **stochastic abstractions** unless externally coupled to ray-tracing engines, and they typically emphasize **protocol-level behavior** over physical-layer accuracy.

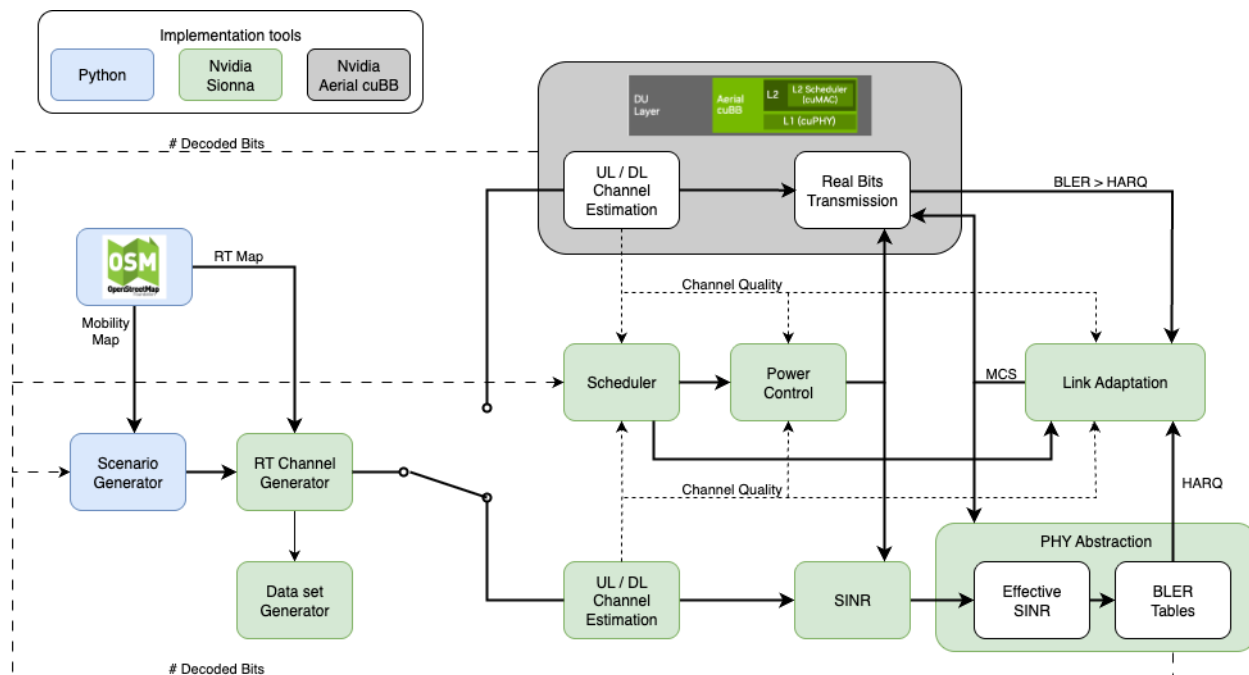


Figure 8. Y-Twin system architecture.

Advancement introduced by the Y-Twin concept beyond state of the art.

1. **Sionna-powered realism.** By building directly on **Sionna RT**, Y-Twin incorporates **site-specific blockage and ray-tracing fidelity**, creating channels that reflect the actual urban geometry of Sinchon. This goes beyond the stochastic approximations common in ns-3 or Simu5G.
2. **3GPP compliance.** Channel outputs are **calibrated using TR 38.901 as reference**, ensuring alignment with industry-accepted statistical baselines. This hybrid approach blends deterministic detail with standardized reproducibility.
3. **5QI-driven service modeling.** Unlike industry site twins, which mostly consider aggregate throughput, Y-Twin embeds **5QI traffic definitions** directly into simulation, enabling service-aware evaluations of scheduling and beamforming.
4. **PHY/MAC observability.** Y-Twin uniquely reports **per-user decoded bits**, not just coverage or signal-to-Interference-plus-noise ratio (SINR). This is crucial for training and benchmarking ML-based PHY/MAC algorithms such as neural receivers or RL schedulers.
5. **Research-focused flexibility.** Whereas industrial twins are tied to operational workflows, Y-Twin is explicitly designed for **experimentation**. This way, these modules can be swapped, scenarios replayed, and new algorithms tested systematically in a controlled environment.

Closing ideas about the Y-Twin concept:

Current DT platforms make significant contributions to deployment, assurance, and planning. However, they do not fully address the needs of **researchers developing AI-native RAN algorithms**. **Y-Twin fills this gap** by offering:

- **site-specific realism** (Sionna ray tracing + 38.901 calibration);
- **service-aware traffic modeling** (via 5QI);
- **decoded-bit observability** (for ML/RL training); and
- **flexible experimentation loops** (replayable, reconfigurable scenarios).

The overall structure of Y-Twin, incorporating the aforementioned functionalities, is illustrated in Figure 8. These strengths make Y-Twin a **unique and powerful platform** for enabling the AI-native

RAN vision, complementing existing industrial and academic efforts while focusing specifically on **data-driven PHY/MAC innovations**.

4. Automation and failure recovery

Beyond performance optimization, 6G networks must ensure resilience, continuity, and autonomous operation under increasingly complex and distributed conditions. This section outlines the planned research directions that explore how AI can enable network self-healing, fault prediction, and intelligent coordination across domains. With AI-based self-healing mechanisms, anomalies can be automatically identified and rectified without human intervention, by recognizing fault patterns and corresponding corrective actions. Similarly, with the use of ML models, historical data can be analyzed to predict equipment failures or network disruptions, scheduling corrective maintenance activities proactively. Finally, AI can also be used to optimize resource allocation dynamically across different network domains, such as RAN, transport networks, and core networks, enabling to achieve the target end-to-end key performance indicators. In this section, these perspectives are complemented by a review of the state of the art in automation, reliability mechanisms, and cross-layer AI integration in the form of four contributions, which are detailed in the following paragraphs.

Contribution 4.1: “AI-based application function for RAN and Core optimizations” considers the extension of the NWDAF, a key component of the 5G network architecture that gathers data from different parts of the network, to collect information from the RAN via the RIC. Based on this information, new AI-based network functions will have the chance to interact with the NWDAF and collect NF resources to anticipate actions at the RAN and Core, aiming at ensuring QoS requirements and prevent congestion end-to-end in the whole mobile network.

Contribution 4.2: “Automated RAN & Core management and Network Failure Recovery” considers the creation of a real-time virtual representation of the RAN and Core networks, also known as Digital Twin (DT). This way, it is possible to visualize and monitor the operation of the mobile network end-to-end, without the necessity of interacting with the physical system directly. This way, if e.g., the DT detects malfunctions or critical issues within the network, it can trigger alerts for immediate attention to take corrective actions to minimize downtimes and service disruption.

Contribution 4.3: “Intelligent Mobility Management in Next-Gen Networks” explores the challenges of resilience and dependability in distributed NextG networks. This contribution focuses on the proactive and smart management of network failures, in which different approaches for cross-domain coordination, semantic monitoring, and adaptive network behaviour are presented. Here, roaming is also considered to support not only user mobility, but also to provide a resilience mechanism that enables automated traffic redirection in response to internal network degradation, ensuring service continuity across heterogeneous domains.

Finally, Contribution 4.4: “Cross-domain AI/ML Coordination in RAN and CN” discusses the challenges behind the integration of AI-based capabilities across all network domains, including the RAN, Core, device, and management layers. Since each of these network domains have unique characteristics and requirements, AI/ML integration on them introduces additional challenges that are also briefly discussed. The optimization that is considered between RAN and Core domains is done using integrated analytics for inference coordination and integrated reconfiguration, enabling a better coordinated decision-making and resource optimization between RAN and Core parts.

4.1. AI-based Application Function for RAN and Core optimizations

In 3GPP Rel-17, more specifically in TS 23.501 [TS23501] and TS 23.288 [TS23288], the System Aspects Working Group 2 (SA2) has defined a new component of the 5G Core which was named NWDAF, as illustrated in Figure 9, whose aim is to collect and analyze data from various parts of the mobile network to provide insights and predictions that can improve network operations, management, and efficiency. To achieve this goal, the NWDAF has the possibility to subscribe to

NF for receiving notifications when certain values in NFs are exceeded. This way, NWDAF can keep track of the network and its reliability, providing an interface to expose the collected data to operations and maintenance (O&M) applications. Based on this analysis, it can be concluded that the NWDAF is the natural NF in the next generation (NextG) architecture for collecting and analyzing data statistics that AI/ML-based algorithms need to perform the required training to perform the specific target actions.

4.1.1. Research Objectives

A proposed solution to increase reliability can be based on enhancing the role of AI-driven network analytics by extending NWDAF to support more granular remote monitoring, such as load, energy consumption, and round-trip times (RTTs). In this regard, the aim is to increase reliability by using these insights to dynamically update NF profiles in the Network Repository Function (NRF). Each NF has its own profile to describe their capabilities and supported features, including the service endpoints where the NF would be receiving the requests from other NFs. Planned studies in this topic aim at refining real-time data collection and policy-driven adaptation. The section reviews current practices in network data analytics, NWDAF evolution, and integration of AI/ML-based applications in the orchestration of NF in the NextG Core.

The objective of this contribution is to extend NWDAF functionalities to enable the interaction and collection of information from RAN through RAN intelligent controller (RIC). In addition to extending the NWDAF, the objective of this activity includes the design a new application function (AF) that includes AI functionality and will interact with NWDAF to collect NF resource information and anticipate some actions in case of congestion or quality of service (QoS) requirements.

The new AF will include interface with e.g. the User Plane Function (UPF), to measure load, energy consumption, among other relevant key performance indicators (KPIs). The new AF will also collect RTT from the base station, as well as other Key Performance Indicators (KPIs) such as load, energy consumption in the different parts of the RAN. The NWDAF will update the existing NF profile into NRF based on the KPI so the NF profile will assign different priority values depending on whether the NF is preferred for energy saving or reliability.

The proposed 6GARROW AI Network Function will perform intra-network data collection and sharing. The AF will collect network analytics from NWDAF to interact with network slice management (NSM) and policy control function (PCF) for changing network policies to improve resource usage and energy saving orchestrate and scale network functions in distributed architectures using AI.

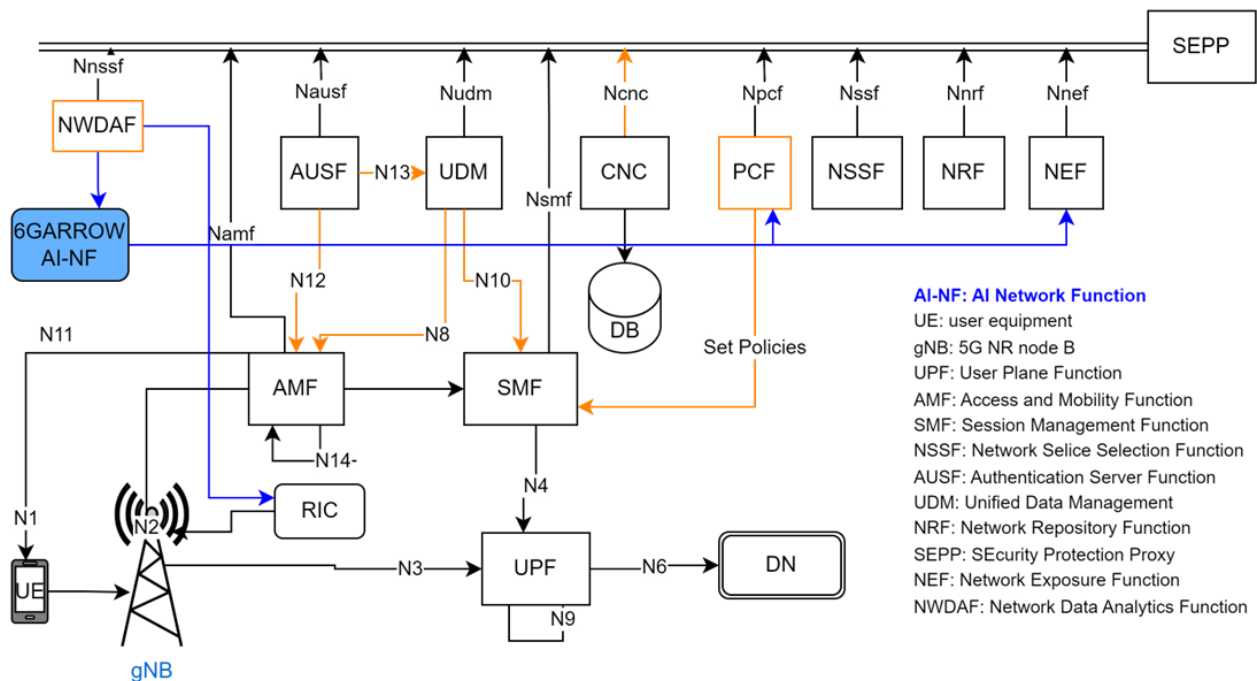


Figure 9. 6G architecture with AI-augmented network function e.g. NWDAF.

4.1.2. Relevant State of the Art

The usage of software defined networks (SDN) controllers have been considered to dynamically reconfigure the network resources. Initially SDN has been used to manage switches and routers in fixed networks, but similar solution has been proposed to manage also mobile backhaul and Radio modules [CPV+18]. The use of Digital-Twin-enhanced (DT-enhanced) AI/ML-based functionalities have also been proposed to monitor and identify system failures to anticipate and launch some corrective actions [HSG+25]. Currently, an end-to-end AI-based orchestrator that has visibility of the resources in different subsystems is being considered. The open challenge with this concept consists of the implementation of suitable discovery mechanism, which is currently not available. This orchestrator should have complete view of the end-to-end infrastructure, to get information about the whole network components and find the most convenient way to interact and manage them. The design of the proposed AI orchestrator work builds upon existing approaches such as SDN with standard interfaces, such as the network configuration protocol (NETCONF), which should be supported by network switches and the different RAN components of the NextG mobile network.

4.2. Automated RAN & Core management and Network Failure Recovery

The development of virtualization platforms that provide execution environments such as virtual machines (VMs), namespaces or containers, such as Docker containers or Kubernetes pods, allow sharing the physical resources across multiple software applications or services. These virtualized environments provide resilience within a single platform in way that, if a NF running in one VM or container crashes, the virtualization platform automatically launches another container with the same network function recreated on that container. The challenge behind this approach is that, since mobile networks consist of several subsystems such as RAN, Transport and Core, each of them with its own platform, an integrated end-to-end management and network recovery module to interact with these sub-systems is required, ensuring the corresponding interoperability needs.

4.2.1. Research Objectives

The standard mechanism to improve resiliency is based on virtualized platforms such as Kubernetes, where different network functions are deployed on pods that consist of virtualized containers that host the NFs and will be re-established in case of failure. The downside of using pods is that, in case of a pod crash event, the new pod will initiate a new NF instance which means that the information of the NF that was running in the old pod is lost. This information consists of active sessions that will be terminated, and devices would have to reestablish a new session with the NF running in the new pod. Thus, a set of persistent state information needs to remain across reboots of network functions running in virtualized platforms.

In addition to virtualized deployments of NF that will be recreated in case of failure, the usage of duplicate connectivity between RAN and multiple instances of core network functions sharing UE state information will be analyzed. This solution would be based on the reliability-aware architectural extensions, including the enhancement of SET-ID defined as part of 3GPP Rel-17 [TS 29.571] through AI-based connectivity and end-to-end reliability prediction. In 3GPP, the NF SET-ID is defined as a logical identifier used to group related NFs together within a 5G system and, through this feature, the sharing of resources can be enabled. An NF Set-ID can be used to allow any NF to handle the requests from other NFs to provide stateless concurrency within the NFs that are part of the same Set-ID. The deployment of DTs for multi-instance monitoring and the support for high-availability inter-public land mobile network (PLMN) mobility with 5G eSIM capabilities is another topic that deserves to be explored. For this purpose, the current landscape of reliability mechanisms, DT deployment in networks, and inter-domain mobility solutions must be known in detail as part of the state of the art review.

Besides the usage of duplicate instances of the core, each of them installed on virtualized environments to guarantee automated recovery as illustrated in Figure 10, the design of intelligent fault management and recovery will be considered. The proposed work will study distributed core architectures under QoS constraints, with a focus on proactive resilience via failure prediction and anomaly detection. The section surveys the current methodologies for resilient core network design, predictive fault management, and AI/ML-assisted recovery mechanisms.

Finally, the usage of AI/ML coordination across the RAN and core domains would be designed based on the concept of AI oracle for managing different system resources. Planned research includes joint data analytics, model lifecycle management, and inference orchestration for coordinated network adaptation. In this regard, the SotA in terms of cross-domain orchestration frameworks, AI pipeline management, and integrated control loops for 6G environments must be also reviewed in detail.

The enhancement of SET-ID [TS 29.571] with AI connectivity analysis and end-to-end reliability prediction is another objective of this contribution. The SET-ID together with NextG DT will be designed for cloud monitoring of multiple instances on-prem and cloud Next5G/6G systems. The DT will monitor the resource allocation from all the NextG deployments from centralized monitoring system, which will also include a DT representation of each instance of NextG system [HSG+25]. The DT that is envisioned will also have the chance to check status of the RAN and Core, reporting performance metrics for the proposed improvements in the management of the network, or triggering alarms when detecting some malfunctioning that will deserve the implementation of self-healing mechanisms.

This objective will be achieved by designing an AI-based orchestrator as an additional module to the 3GPP standard NWDAF to interact with distributed AI agents handling their own subsystem e.g. RAN, Transport and Core. A suitable standard protocol, such as NETCONF for the E2E AI orchestrator, will be considered to interact with the subsystem AI agent. After that, SET-ID concept with the AI orchestrator will be designed for managing deployment on premise or on the cloud to provide seamless network failure recovery. An analysis of DT for collecting data from network subsystems to train the AI orchestrator, for defining recovery actions, also needs to be performed.

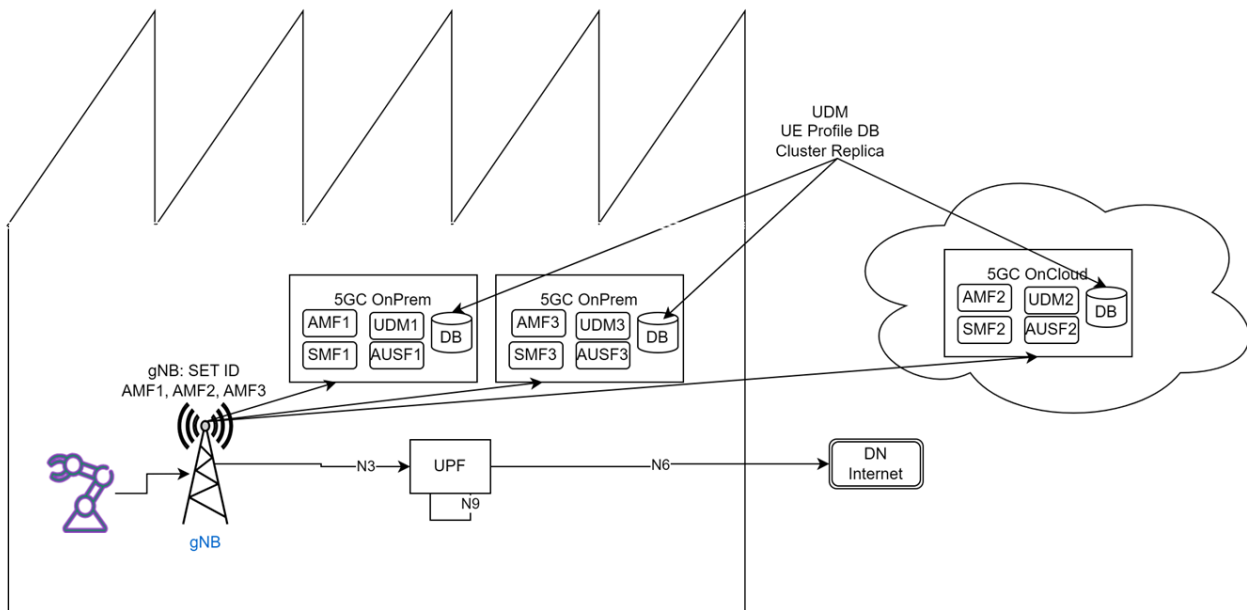


Figure 10. RAN, Transport and Core management and E2E recovery orchestrator.

4.2.2. Relevant State of the Art

The 3GPP Rel-17 [TS 29.571] has defined SET-ID as solution for high availability based on the reliability-aware architectural extensions. The SET-ID, ing of a logical identifier to group multiple instances of the same network function, can help in providing load balancing and high availability features without session disconnection compared to 4G mobility management entity MME pooling and S1-Flex.

The usage of DTs has been considered in different scenarios to reflect the functioning of different systems. DT has been proposed to mirror the status of the mobile networks. The open challenge in this regard is to build a generic DT that provides a virtual image of RAN, transport and core for the AI orchestrator to have complete visibility and anticipate to failure events and be prepared with recovery actions. The DT has been proposed to monitor the resource allocation from all the network components from centralized s and include a DT representation of each instance of the mobile infrastructure [HSG+25].

4.3. Intelligent Mobility Management in Next-Gen Networks

Next-generation mobile networks including the 5GC and the future 6G systems present a new set of problems concerning dependability and resilience based on their distributed architecture. Unlike centralized systems, core network operations are scattered among several administrative and technological spheres. This distributed character increases the surface exposed to partial failures, local overloads, configuration errors, and other unpredictable events even while it improves scalability and flexibility. These issues may compromise end-to-end network performance and maybe lead to QoS violations impacting consumers and applications. Avoiding these negative consequences depends on designing intelligent systems able to constantly monitor network status, predict future hazards, and react in time to stop service disturbance. Such systems must include mechanisms for adaptive behavior that transcend basic static rules and react to changing dynamic network conditions. This work will explore these mechanisms with an eye toward how prediction, observation, and automated control might be merged to support proactive management of important network events in developing 5GC and 6G infrastructure.

4.3.1. Research Objectives

One crucial instance of these circumstances is the activation of roaming between public and private networks, exploited as a resilience mechanism in addition to being a support to mobility. Although roaming usually corresponds with user movement across network areas or operators, in a resilient 5GC or 6G system it can be triggered by internal conditions implying a possible QoS degradation. Should a private network domain exhibit early indicator of instability, such as elevated latency, CPU or memory saturation on edge nodes or aberrant signaling behavior, the system may decide to reroute sessions or data flows toward a public network domain providing more stable conditions, when data privacy requirements allow it. Likewise, roaming can start when a domain loses the ability to provide necessary security operations, including entity authentication across domains or policy implementation. Under such circumstances, functionalities as the security edge protection proxy (SEPP) become indispensable to guarantee safe interconnection and to verify confidence in communications between operators before traffic is allowed to flow across domains. Designing strong architectures that preserve QoS while preserving operational flexibility across heterogeneous domains in the 5GC and 6GC context depends on a clear knowledge of how such transitions are triggered, tracked, and regulated.

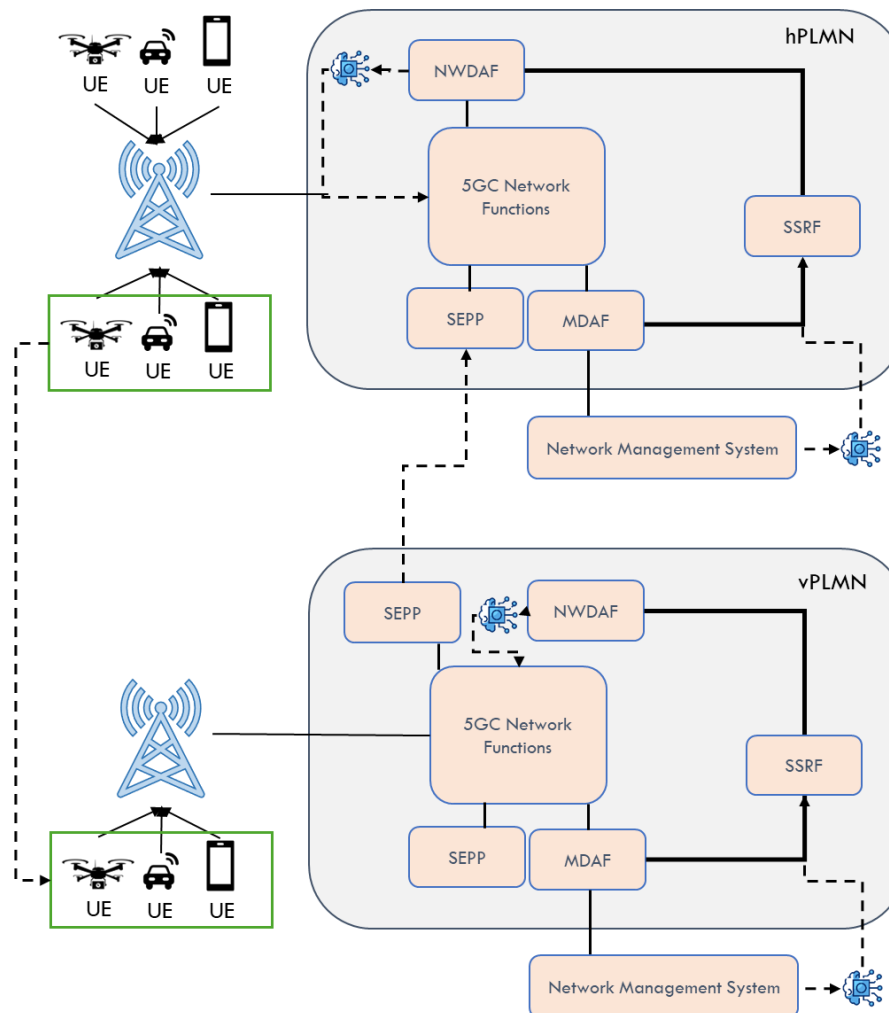


Figure 11. Inter-PLMN with SSRF

Multiple systems must cooperate across several layers of the central network if we are to enable such smart and automated decisions. By compiling and evaluating structured and unstructured data from many network segments and layers, intelligent data processing systems including the SSRF [6GARROW-D21] can be centrally important (Figure 11). The SSRF is a system that collects and combines data from different parts of the 5GC. It uses smart algorithms and language models to

change complex network information into a clear format that both humans and machines can understand, helping operators manage the network better, detect problems earlier, and support future 6G functions like devices communicating with each other without human help. While supporting contextual knowledge of current network conditions, this semantic abstraction helps the system detect developing patterns suggesting possible anomalies or service degradation. This information lets the system suggest or independently implement mitigating strategies including traffic reallocation, virtualized network function reallocation, or movement of operations to more stable environments. Concurrently, other modules could verify compliance with expected behavior or identify minor deviations in security policies that might grow over time, so acting as short-term prediction of network evolution. These elements taken together create an adaptive and distributed defense layer able to act locally and globally to maintain service continuity and maximize performance in different environments. Shaping next-generation resilience strategies in line with the 6G vision, where networks are self-adaptive, secure, and capable of intelligent self-recovery with minimum human intervention, depends on investigating and integrating these mechanisms.

4.3.2. Relevant State of the Art

New difficulties arise from the distributed nature of 5GC and upcoming 6G networks. While these networks aim for scalability and adaptability, they pose technical and organizational challenges such as maintaining continuous high-quality service, making autonomous decisions, and preventing service interruptions. Network operations are often managed by multiple administrative domains, which may not always coordinate effectively. This separation increases the risk of configuration errors, traffic congestion, and cascading failures [FM+24]. Smart systems are needed to monitor networks in real-time, detect early warning signs, and act autonomously to prevent user impact [VHM+24].

Roaming, originally designed to maintain user connectivity across locations and operators, has evolved into a resilience mechanism. National roaming agreements allow traffic to be rerouted during network overloads or failures, reducing service interruptions and maintaining call, message, and data continuity [LSC+24]. This concept is increasingly relevant in private networking scenarios, where numerous private networks coexist with public networks, and automated user provisioning is growing. Private networks support high-performance tasks in industrial, hospital, and corporate environments, while public networks serve many users and devices over wide areas. Seamless transitions between private and public networks are critical, as delays or losses can cause major disruptions. Traffic or network slice migration between networks preserves performance under congestion or limited resources [JVS+24]. In these cases, the SEPP ensures secure communication, identity verification, and data protection.

Current 3GPP standards, however, do not support roaming between PLMNs and standalone non-public networks (SNPN), limiting this capability [MFB+22]. Relaxing this constraint and introducing more flexible solutions could improve traffic management, speed up responses to disruptions, and enhance service reliability.

Researchers are exploring automated roaming systems to address these limitations. The SSRF [6GARROW-D21] uses semantic technologies and LLMs to gather data from multiple network segments and make intelligent decisions, including activating roaming or reallocating traffic without human intervention [VHM+24]. Network slicing and automatic roaming rules also enable service migration across network segments, improving performance and reducing delays during high traffic or partial failures [JVS+24].

As 6G approaches, roaming is increasingly essential not just for mobility, but for building resilient, adaptable, and dependable networks. While current standards lag this vision, intelligent systems like SSRF and automatic slice management show strong potential. Updating guidelines to accommodate these technologies is crucial for next-generation networks capable of autonomously handling complexity and disruptions.

4.4. Cross-domain AI/ML Coordination in RAN and CN

6G mobile networks are envisioned to be AI-native, meaning that AI/ML capabilities are integrated across all network domains, including the RAN, CN, device, and management layers. Unlike 5G, where AI-based optimizations have predominantly been siloed within either the RAN or CN, 6G aims to achieve end-to-end intelligence that spans both domains. This vision necessitates breaking the traditional separation between the RAN and CN, enabling cross-domain AI/ML coordination to optimize end-to-end performance, resource efficiency, and user experience throughout the entire network. Specifically, this technology focuses on the following aspects: (1) optimization between RAN and CN domains using integrated analytics, (2) design of data, RAN-CN integrated analytics, and AI model management, and (3) design of inference coordination and integrated reconfiguration.

4.4.1. Research Objectives

Optimization between RAN and CN domains using integrated analytics refers to the ability to perform task optimization in an end-to-end manner. For instance, in network slicing, both RAN and CN slices require harmonized configuration to guarantee end-to-end performance as illustrated in Figure 12. To this end, analytics that incorporate both RAN and CN attributes must be developed. Furthermore, designing a framework for RAN-CN integrated analytics and AI model management is essential to generate insights that support decision-making processes. Since data collected from a single domain is insufficient to realize AI/ML-driven analytics for a comprehensive view of the network, appropriate data features and acquisition mechanisms must be established while addressing privacy concerns.

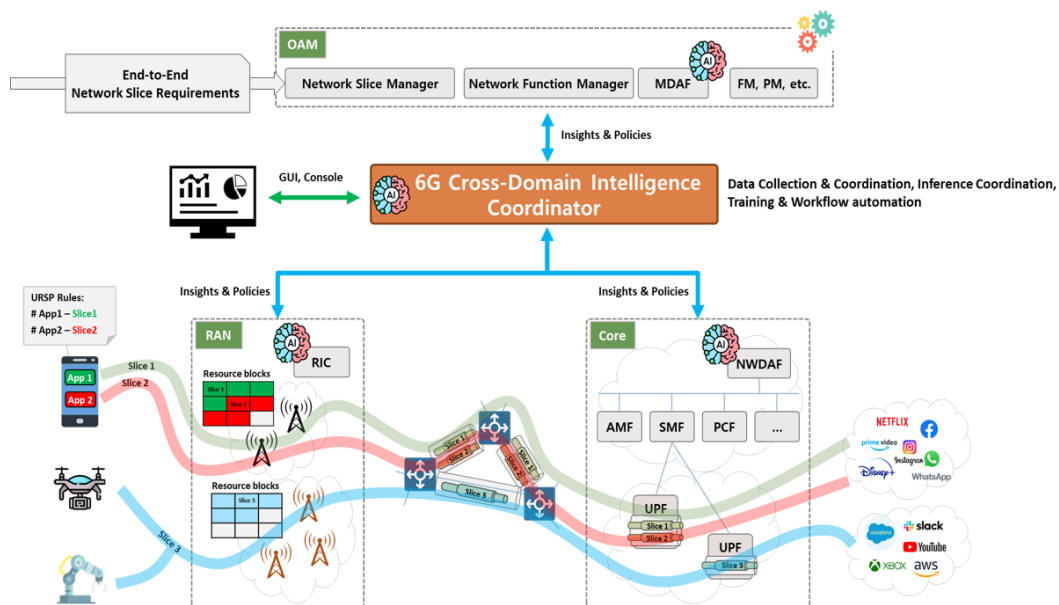


Figure 12. Cross-domain AI/ML coordination for end-to-end network slicing

Once models are trained, they require periodic evaluation to ensure that their analytics outputs remain valid for network configuration tasks. Finally, inference coordination and integrated reconfiguration address the need to detect and mitigate potential conflicts between inferences generated within individual domains that may otherwise degrade overall performance. While RAN and CN inferences provide valuable domain-specific insights (i.e., local optima), they may fail to achieve optimal network-wide outcomes (i.e., global optima). Each inference, therefore, must be assessed in terms of its contribution to overall performance objectives. When negative impact is identified, inference outputs should be adjusted to align with the broader goals of network optimization.

4.4.2. Relevant State of the Art

At present, the RAN and CN domains each employ their own analytics and control functions. In the RAN domain, the RIC within the open RAN architecture manages RAN-specific optimizations [nGRG24], whereas the NWDAF provides analytics for the CN domain [TS23288]. However, these components operate independently with limited coordination, thereby hindering the performance enhancements necessary for 6G mobile networks. To address this limitation, recent advancements have focused on developing integrated RAN-CN AI frameworks that support cross-domain functionality. These frameworks include mechanisms for data sharing, unified management of AI model lifecycles across domains, distributed learning across network segments, and the generation of integrated analytics to enhance end-to-end performance.

The design of an integrated AI/ML architecture that spans both RAN and CN domains is critical to elevating domain-specific AI capabilities to the cross-domain level. In 5G networks, control functions are typically deployed in a domain-isolated manner, resulting in few interfaces for direct communication between the RAN and CN. This architectural separation obstructs efficient end-to-end network operation. To overcome this obstacle, recent discussions have proposed a converged RAN-CN control plane for 6G networks [CHOI22], which leverages a service-based interface (SBI) applicable throughout the entire network, eliminating the boundary between domains.

Beyond architectural convergence, potential solutions for cross-domain AI built on the O-RAN framework have also been introduced [nGRG24]. These include RAN-CN converged functions designed not only to remove functional redundancy but also to reduce unnecessary forwarding overhead between the RAN and CN. Furthermore, enhanced service management and orchestration (SMO)-based cross-domain AI capabilities have been proposed, incorporating elements such as a cross-domain AI control function and full lifecycle management of AI models across domains. In parallel, an alternative approach utilizing the NWDAF functionality defined by 3GPP [GP24] has also been explored. Given NWDAF's capabilities for distributed and decentralized machine learning, it is well positioned to enable cross-domain AI/ML coordination. It can derive CN-related insights from NFs and obtain RAN data via operations, administration, and maintenance (OAM) interfaces.

From the perspective of AI model management in support of cross-domain coordination, standard development organizations (SDOs) such as 3GPP and ETSI have proposed relevant frameworks. In the case of 3GPP, the management data analytics (MDA) function and the associated management data analytics service (MDAS) are designed to process and analyze data across multiple network management domains [TS28104]. Functionally, MDAS serves as the management-plane counterpart to NWDAF. It consumes multi-domain telemetry, including RAN and core KPIs, and produces analytics to support closed-loop automation within orchestration systems. Similarly, ETSI's zero-touch network and service management (ZSM) architecture defines an end-to-end service management domain encompassing multiple management sub-domains [ZSM002]. The ZSM framework facilitates cross-domain data exposure and provides common services, including analytics and AI functionalities, that can be shared across domains. Both 3GPP and ETSI envision AI-as-a-service capabilities in future networks, and this vision remains relevant for 6G. For example, this includes a catalog of deployable AI models for distinct domains, or data services that allow one domain (e.g., the CN) to query RAN data on demand.

Distributed learning techniques, such as federated learning (FL) and split learning (SL), across network segments are also critical for addressing both data privacy concerns and disparities in computational capacity across network segments. These techniques are vital for generating integrated analytics that span both RAN and CN domains in 6G networks. In contrast to centralized AI methods, distributed learning allows training and inference to be conducted collaboratively across domains without transferring raw data. This is particularly beneficial for latency-sensitive applications, data privacy, and real-time responsiveness. For instance, a prediction-assisted federated deep reinforcement learning framework has been proposed to enable RAN and CN agents to jointly optimize slice mobility decisions in real time, thereby reducing service disruption during user mobility across domains [ZM24].

Architecturally, a decentralized collaborative learning system has been introduced in which both FL and SL are simultaneously employed across RAN and CN domains. This system enables joint training of models that enhance core network operation and coordination, all while preserving data confidentiality [MH24]. Additional work has explored how SL can be adapted to 6G scenarios by partitioning neural networks between RAN and edge/CN segments. This approach enables efficient, privacy-aware inference close to the user while offloading computationally intensive tasks to the edge or CN [ZH24]. Furthermore, a hybrid approach known as split federated learning (SFL) has been proposed [HH24]. This technique combines the advantages of both FL and SL, enabling RAN and CN to collaboratively train different portions of a shared model while addressing challenges such as data heterogeneity, convergence stability, and communication overhead. Collectively, these developments emphasize the feasibility and importance of distributed learning as a foundation for tightly coordinated, intelligent behavior across the RAN-CN domains in 6G networks.

Despite the progress demonstrated by the aforementioned efforts, several gaps remain on the path toward fully coordinated RAN-CN AI in 6G. First, the RAN and CN still predominantly operate on distinct datasets. Sharing information across the RAN-CN boundary is complicated by data volume and privacy concerns. A key challenge lies in determining how to expose only the necessary data for effective cross-domain AI/ML coordination, without overloading interfaces or violating regulatory constraints. Although distributed learning techniques such as federated learning can alleviate some of these issues, they are largely focused on model training rather than on real-time data acquisition required for decision-making.

Additionally, ensuring consistency and temporal alignment of data remains an unresolved challenge, as RAN and CN measurements often occur at different timescales and levels of granularity. Interoperability and cross-vendor coordination also pose significant difficulties. Achieving effective cross-domain AI/ML coordination will require integration of solutions from different vendors and functional domains, which raises issues related to standardization of model formats, APIs, and semantic representations [LZ25]. Another architectural challenge concerns the lack of a unified entity responsible for overseeing cross-domain AI/ML coordination. While various frameworks have been proposed by industry projects and SDOs, a unified inference framework that seamlessly spans RAN and CN domains has yet to be established.

5. Conclusion

Deliverable D4.1 has established the foundation for Objective 3: Optimize RAN and Core based on AI/ML solutions of the 6GARROW project. For this purpose, it presented both the planned research directions and the state of the art related to AI-native architectures, resource optimization, and automation mechanisms for future 6G systems.

The analysis confirms that AI/ML-based algorithms for radio resource management and semantic intelligence are both key enablers for achieving energy efficiency, adaptability, and resilience in next-generation mobile networks. At the same time, this deliverable highlights the need for new architectural concepts capable of ensuring trustworthiness and regulatory compliance, notably with respect to the forthcoming EU AI Act. The review also shows that, while significant progress has been made in AI/ML-assisted RAN and Core optimization, challenges remain in cross-domain coordination, scalability of learning mechanisms, and integration of explainable and accountable AI/ML models.

The insights gathered in this deliverable will guide the subsequent stages of the project, particularly in the design, implementation, and validation of AI/ML mechanisms to be addressed in D4.2 and D4.3. These next steps will build upon the identified gaps and research opportunities to deliver demonstrable results on intelligent, energy-aware, and self-managing 6G network operation.

Ultimately, this deliverable provides a strategic and technical baseline for the ambition behind 6GARROW, aiming at positioning Europe at the forefront of trustworthy, AI-native 6G innovation, ensuring that future networks are not only more capable, but also more sustainable and reliable.

6. References

- [6GARROW-D21] 6GARROW, Deliverable D2.1 "6GARROW Scenarios, Use Cases and related KPIs/KVIs," 08/2025.
- [AFP+22] P. Almasan, et al., "Network Digital Twin: Context, Enabling Technologies, and Opportunities," *IEEE Communications Magazine*, vol. 60, no. 11, pp. 22–27, 2022. doi:10.1109/MCOM.001.2200012.
- [AHMA24] A. Al-Ahmadi, "A Convolutional Neural Network for Beam Prediction in 5G and Beyond Massive MIMO Systems," *Journal of Engineering and Applied Sciences*, 11(1), 141–151, 2024.
- [AHME25] Al-Tahmeesschi, Ahmed, et al. "Enhancing Open RAN Digital Twin Through Power Consumption Measurement." *arXiv preprint arXiv:2507.00928* (2025).
- [CHOI22] Choi, Jinho, et al. "RAN-CN converged control-plane for 6G cellular networks." *GLOBECOM 2022-2022 IEEE Global Communications Conference*. IEEE, 2022.
- [CPV+18] "SDN-Based UPF for Mobile Backhaul Network Slicing", Jose Costa-Requena; Aapo Poutanen; Serdar Vural; George Kamel; Chris Clark; Shourov Kumar Roy, E uCNC 2018 (<https://ieeexplore.ieee.org/document/8442795>)
- [CSM+25] "Interoperability and Coexistence of 6G Semantic, Goal-Oriented, and Legacy Systems." *Foundations of Semantic Communication Networks* (2025): 125-140.
- [ETSI-SDN] "Virtual EPC with SDN Function in Mobile Backhaul Networks", ETSI PoC 2015, <https://nfvwiki.etsi.org/index.php>
- [EUAIAct] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act) (Text with EEA relevance), available at <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
- [FIL125] B. De Filippo, et al., "Uplink OFDM Channel Prediction with Hybrid CNN-LSTM for 6G Non-Terrestrial Networks," *arXiv:2502.09326*, 2025.
- [FM+24] Fatema Bannat Wala, Mariam Kiran, "5G Network Security Practices: An Overview and Survey", *arXiv:2401.14350v1*.
- [GAN25] J. Gan, H. Kou, G. Yang, H. Zhang, Z. Cao and W. Xu, "Joint Sleep Control and Energy Sharing Strategy With Deep Reinforcement Learning in Green Ultra-Dense Networks," in *IEEE Transactions on Green Communications and Networking*
- [GAO25] D. Gao *et al.*, "Joint Load Adjustment and Sleep Management for Virtualized gNBs in Computing Power Networks," in *IEEE Transactions on Wireless Communications*, vol. 24, no. 3, pp. 2067-2082, March 2025
- [GP24] Gkonis, Panagiotis K., et al. "Leveraging network data analytics function and machine learning for data collection, resource optimization, security and privacy in 6G networks." *IEEE access* 12 (2024): 21320-21336.
- [HEXAX+23] Deliverable D4.3 AI-driven communication & computation co-design: final solutions, Hexa-X Project, 2023, available at https://hexa-x.eu/wp-content/uploads/2023/05/Hexa-X_D4.3_v1.0.pdf
- [HH24] Hafi, Houda, et al. "Split federated learning for 6G enabled-networks: Requirements, challenges, and future directions." *IEEE Access* 12 (2024): 9890-9930.
- [HMC+24] Hello, Nour, et al. "Optimizing RIS Impairments through Semantic Communication." *arXiv preprint arXiv:2410.08155* (2024).
- [HSG+25] "Digital Twins of Industrial and 6G Systems: Enablers Towards Situational Awareness" Armin Hadziaganovic, Joachim Sachs, James Gross, Damir Hamidovic, Mahin Ahmed, Raheeb Muzaffar, Andreas Springer, Hans-Peter Bernhard (<https://silicon-austria-labs.elsevierpure.com/en/publications/digital-twins-of-industrial-and-6g-systems-enablers-towards-situa>)
- [JAYA24] N. Jayaweera, et al., "5G-Advanced AI/ML Beam Management: Performance Evaluation with Integrated ML Models," *arXiv:2404.15326*, 2024.

- [JIAN24] C. Jiang, et al., "AI for CSI Prediction in 5G-Advanced and Beyond," arXiv:2504.12571, 2024.
- [JKIM25] J. Kim, et al., "Masked Transformer for Robust CSI Prediction with Incomplete Pilots," IEEE Trans. Wireless Commun., vol. 24, no. 5, pp. 3210–3223, 2025.
- [JVS+24] Joao P. Ferreira, Vinicius C. Ferreira, Sergio L. Nogueira, et al., "A Flexible Infrastructure-Sharing 5G Network Architecture Based on Network Slicing and Roaming", arXiv:2411.15505v1.
- [KART24] R. M. Karthik, et al., "Machine Learning (ML)-assisted Beam Management in millimeter (mm)Wave Distributed Multiple Input Multiple Output (D-MIMO) systems," arXiv:2401.05422, 2024.
- [KARU24] S. Karunasena, et al., "Deep Learning based Computer-vision for Enhanced Beamforming," arXiv:2412.03073, 2024.
- [KORA25] I. Koralege, et al., "A Deep-Unfolding Approach to RIS Phase Shift Optimization Via Transformer-Based Channel Prediction," arXiv:2502.18280, 2025.
- [KSS+24] A. Kaushik et al., "Toward Integrated Sensing and Communications for 6G: Key Enabling Technologies, Standardization, and Challenges," in IEEE Communications Standards Magazine, vol. 8, no. 2, pp. 52-59, June 2024, doi: 10.1109/MCOMSTD.0007.2300043.
- [LC24] Liu, Yuanwen, and Bruno Clerckx. "Rate-Splitting Multiple Access for Coexistence of Semantic and Bit Communications." arXiv preprint arXiv:2409.10314 (2024).
- [LIAN24] J. Liang, et al., "Aligning Beam with Imbalanced Multi-modality: A Generative Federated Learning Approach," arXiv:2504.14835, 2024.
- [LKD+23] X. Lin, et al., "6G Digital Twin Networks: From Theory to Practice," IEEE Communications Magazine, vol. 61, no. 11, pp. 72–78, Nov. 2023. doi:10.1109/MCOM.001.2200830
- [LSC+24] Lotte Weedage, Syllas R. C. Magalhães, Clara Stegehuis, et al., "On the Resilience of Cellular Networks: How Can National Roaming Help?", IEEE Transactions on Network and Service Management, Volume 21, Issue 2, DOI: 10.1109/TNSM.2024.3352168.
- [LSX+25] Lam Dinh, Sihem Cherrared, Xiaofeng Huang, and Fabrice Guillemin, "Towards End-to-End Network Intent Management with Large Language Models", arXiv 2504.135.
- [LZ24] Li, Zexu, et al. "The Architecture of AI and Communication Integration towards 6G: An O-RAN Evolution." Proceedings of the 30th Annual International Conference on Mobile Computing and Networking. 2024.
- [LZ25] Li, Zexu, et al. "Evolving Towards Artificial-Intelligence-Driven Sixth-Generation Mobile Networks: An End-to-End Framework, Key Technologies, and Opportunities." Applied Sciences 15.6 (2025): 2920.
- [MARE24] L. Marengo, et al., "Machine-learning-aided method for optimizing beam selection and its update time," Nature Scientific Reports, 2024.
- [MF25] Merluzzi, Mattia, and Miltiadis C. Filippou. "Goal-oriented Spectrum Sharing: Trading Edge Inference Power for Data Streaming Performance." arXiv preprint arXiv:2503.11552 (2025).
- [MFB+22] Marius Corici, Fabian Eichhorn, Bjoern Riemer, et. al., "Extended Coverage without Roaming for beyond 5G Non-Public Networks", GLOBECOM 2022 - 2022 IEEE Global Communications Conference, DOI: 10.1109/GLOBECOM48099.2022.10001566.
- [MFB+24] Merluzzi, Mattia, et al. "6G goal-oriented communications: How to coexist with legacy systems?." Telecom. Vol. 5. No. 1. MDPI, 2024.
- [MH24] Hossain, Mohammad Arif, et al. "A distributed collaborative learning approach in 5G+ core networks." IEEE Network (2023).
- [MLG+23] Mu, X., Liu, Y., Guo, L., & Al-Dhahir, N. (2023). Heterogeneous semantic and bit communications: a semi-NOMA scheme. IEEE Journal on Selected Areas in Communications, 41(1), 155-169.
- [MMR+25] Maxime Elkael, Michele Polese, Reshma Prasad, "ALLSTaR: Automated LLM-Driven Scheduler Generation and Testing for Intent-Based RAN", arXiv:2505.18389v2.
- [MPY+25] Md Arafat Habib, Pedro Enrique Iturria Rivera, Yigit Ozcan, et al., "Harnessing the Power of LLMs, Informers and Decision Transformers for Intent-driven RAN Management in 6G", arXiv:2505.01841v1.

- [MUECK+25] European Digital Regulations (Intelligent Systems Reference Library, 265, Band 265) Gebundene Ausgabe, Springer Nature, ISBN 978-3031808081, 25. February 2025
- [NGJCAS24] Next G Alliance Report: Next G Alliance Report: Channel Measure-ments and Modeling for Joint/Integrated Communication and Sensing, as well as 7-24 GHz Communication, July 2024. https://nextgalliance.org/white_papers/channel-measurements-and-modeling-for-joint-integrated-communication-and-sensing-as-well-as-7-24-ghz-communication/
- [nGRG24] Z. Li et al., "Research report on cross-domain AI", RR- 2024-02, O-RAN Alliance Next Generation Research Group (nGRG), 2024
- [PBD+23] M. Polese, et al., "CoIO-RAN: Developing Machine Learning-Based xApps for Open RAN Closed-Loop Control on Programmable Experimental Platforms," IEEE Transactions on Mobile Computing, vol. 22, no. 10, pp. 5787–5800, Oct. 2023. doi:10.1109/TMC.2022.3188013
- [QIAN22] Q. Li et al., 6G Cloud Native System: Vision, Challenges, Architecture Framework and Enabling Technologies, IEEE Access, 2022, available at <https://ieeexplore.ieee.org/document/9882121>
- [RAN-SDN] "SDN-Enabled THz Wireless X-Haul for B5G", Jose Costa-Requena; Abraham Afriyie; Chartsias Panteleimon Konstantinos; Karasoula Eleni; Kritharidis Dimitrios; Nicola Carapellese, EuCNC 2021 (<https://ieeexplore.ieee.org/document/9482494>)
- [SDN-CTRL] Botez, R., Costa-Requena, J., Ivanciu, I.-A., Strautiu, V., & Dobrota, V. (2021). Sdn-based network slicing mechanism for a scalable 4g/5g core network: A kubernetes approach. Sensors, 21(11), Article 3773. <https://doi.org/10.3390/s21113773>
- [TR38901] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz (Release 14)," 3GPP TR 38.901 V14.3.0, Sept. 2017. [Online]. Available: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3173>
- [TS 29.571] Common Data Types for Service Based Interfaces; Stage 3 (3GPP TS 29.571)
- [TS22.137] 3GPP, TS 22.137, "Integrated Sensing and Communication", Technical Specification, Release 19, 2024. <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4198>
- [TS23.288] Architecture enhancements for 5G System (5GS) to support network data analytics services (3GPP TS 23.288 version 17.4.0 Release 17)
- [TS23288] 3GPP SA2, TS 23.288, "Architecture enhancements for 5G System (5GS) to support network data analytics services", v19.2.0 (2025-03)
- [TS23501] 3GPP, "System Architecture for the 5G System (5GS); Stage 2," TS 23.501. [Online.] Available: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3149>
- [TS23501] 3GPP, "System Architecture for the 5G System (5GS); Stage 2,
- [TS28104] 3GPP SA5, TS 28.104, "Management and orchestration; Management Data Analytics (MDA).", V19.1.0, 2025.
- [VHM+24] Viktoria Vomhoff, Hyunseok Daniel Jang, Matteo Varvello, et al., "Challenges and Opportunities for Global Cellular Connectivity", arXiv:2411.19706v1.
- [WANG24] X. Wang et al. "Multi-User Beamforming with Deep Reinforcement Learning in Sensing-Aided Communication." arXiv:2505.05956, 2024.
- [WANG25] Y. Wang, et al., "Sequence-to-Sequence LSTM for Robust CSI Prediction in Massive MIMO Systems," IEEE Trans. Wireless Commun., vol. 23, no. 4, pp. 2456–2468, 2025.
- [XLIU25] X. Liu, et al., "Research on channel estimation based on joint perception and deep enhancement learning in complex communication scenarios," PeerJ Comput. Sci., 2025.
- [YCT+25] Yao Wang, Chungang Yang, Tong Li, et al., "A Survey on Intent-Driven End-to-End 6G Mobile Communication System", Published in: IEEE Communications Surveys & Tutorials, DOI: 10.1109/COMST.2025.3575041.
- [YOU25] C. You, X. He, Y. Sun, G. Feng and T. Q. S. Quek, "GreenRAN: A Channel-Aware Green O-RAN Framework for NextG Mobile Systems," IEEE INFOCOM 2025 - IEEE Conference on Computer Communications, London, United Kingdom, 2025, pp. 1-10

- [ZH24] Lin, Zheng, et al. "Split learning in 6G edge networks." IEEE Wireless Communications (2024).
- [ZM24] Ming, Zhao, Hao Yu, and Tarik Taleb. "Federated Deep Reinforcement Learning for Prediction-Based Network Slice Mobility in 6G Mobile Networks." IEEE Transactions on Mobile Computing (2024).
- [ZSM002] ETSI GS ZSM 002, "Zero-touch network and Service Management (ZSM); Reference Architecture.", V1.1.1, 2019
- [ZZLI25] Z. Li, et al., "Bridging the Modality Gap: Enhancing Channel Prediction with Semantically Aligned LLMs and Knowledge Distillation," arXiv:2505.12729, 2025.